

Mid-Depth Retrieval Collapse in T5 Encoders: A Layerwise Case Study on Medieval Latin Witnesses

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Abstract

Practitioners often read pre-trained encoders at intermediate layers, diagnose them with an anisotropy statistic, and correct them with a post-hoc projection. These steps assume that the statistic tracks retrieval quality. We test that assumption layer by layer on 1,705 medieval Latin canon-law witnesses, whose equivalence labels come from a scholarly edition. In the middle layers of three T5 encoders, cosine retrieval falls to or near chance (AUROC 0.50–0.65), while three embedding-trained models stay above 0.80 at every layer. Mean pairwise cosine gives no warning. Top-PC share, the first principal component’s variance share, flags all 26 collapsed layers with one false alarm in 100, but orders layers only weakly within each group. A top- D projection fit on training vectors lifts every layer to AUROC 0.96–0.99 and cuts the cross-model spread in routing accuracy from 39.3 to 3.3 points. The repair matches, but does not exceed, a character n-gram reference. Contrastive fine-tuning of LaTa improves its last layer but not its collapsed middle layers. We test whether a few massive, passage-varying residual coordinates cause the collapse, and whether T5 pretraining or the missing embedding objective produces it [Pending: E1–E3, P2x2: one-sentence result]. We release the frozen testbed and split, the evaluation code, and lexical reference scores.

1 Introduction

Pre-trained encoders are a standard representation for semantic matching in text-reuse and digital-humanities pipelines (e.g., Manjavacas et al., 2019; Schelb et al., 2026; Reichbauer et al., 2026). Practitioners take a model off the shelf and often read an intermediate layer, since intermediate layers can outperform the final one (Skean et al., 2025). They diagnose the embedding space with an anisotropy statistic, usually the mean cosine between passages, adapted from the word-level statistic of Ethayarajh

(2019). They then apply a post-hoc correction, such as removing the top principal directions (Mu and Viswanath, 2018) or whitening (Su et al., 2021). The last two steps assume that the statistic tracks retrieval quality: a layer the statistic calls healthy should retrieve well, and a layer it flags should gain from the correction. We test that assumption layer by layer.

Medieval canon law gives a controlled case for this test. Before the twelfth century it survives in many compilations, whose witnesses differ in spelling, wording, and fragment boundaries. Matching a witness to its source by hand is labor-intensive expert work, because scribes often omitted or misnamed the source. The Carolingian Canon Law project (CCL; Firey, 2009) has already made these decisions for 1,705 witnesses of 840 sources. Its labels are expert-assigned by the project and checked by collation in its parallel display, so they are independent of any model. A character n-gram reference retrieves at least as well as every pre-trained encoder configuration we test, so the equivalence signal is present in the surface text. A retrieval failure therefore points to the representation rather than to the data. Latin also has dedicated encoders in two architectures. The T5 models LaTa and PhilTa have encoder-only siblings, LaBERTa and PhilBERTa, trained by the same authors on the same data (Riemenschneider and Frank, 2023). Latin BERT (Bamman and Burns, 2020) is a further encoder-only model. This allows a matched comparison of architectures (Section 6).

The layerwise view shows a failure that the usual diagnostic misses. In the middle layers of the three T5 encoders in our panel, cosine retrieval falls to or near chance: pairwise AUROC at the worst layer is 0.50 for LaTa, 0.54 for PhilTa, and 0.65 for mT5-base. The three embedding-trained models stay above 0.80 at every layer. Mean pairwise cosine gives no warning. It is highest in the models that retrieve best, and within the T5 encoders it is lower

at the collapsed layers than at the healthy ones. Variance statistics do better. A top-PC share of 0.6 or more marks all 26 collapsed layers, with one false alarm in 100 model-layers. Within a regime, however, top-PC share and effective rank order layers only weakly (Section 4). A top- D projection fit on training vectors (ABTT; [Mu and Viswanath, 2018](#)), with no parameter update, lifts every layer of every model to AUROC 0.96 to 0.99. It also compresses the spread in routing accuracy across models from 39.3 to 3.3 points. The repair matches the n-gram reference and does not exceed it. Contrastive fine-tuning raises LaTa’s last layer from 0.938 to 0.984 AUROC, yet its layers 2 to 11 stay at 0.50 to 0.57.

Prior work shows that a few residual coordinates of very large magnitude can dominate cosine similarity ([Timkey and van Schijndel, 2021](#)). In decoder language models they persist through the middle layers ([Sun et al., 2024](#)), and in T5 encoders they grow with depth ([Zhao et al., 2025](#)). We test whether mean pooling turns these coordinates into values that vary from passage to passage and so drown out content (Section 5). We also test whether T5 pretraining or the missing embedding objective produces the collapse (Section 6).

We measure the geometry on two tasks. Pairwise retrieval (Task A) scores every pair of witnesses and tests ranking directly. Open-set routing (Task B) files each witness under a known source or flags it as new, which adds a threshold decision. The embedding-trained models show a second regime. In them every passage shares a large mean vector, which raises every cosine (mean pairwise cosine 0.59 to 0.97), while variance stays spread over many directions. These models rank well, yet the projection still raises the share of witnesses LaBSE files correctly as known or new from 83.3 to 90.8 percent. Section 7 tests whether this gain goes beyond ranking.

This paper makes four contributions. (1) We give the first per-layer retrieval profile of T5 encoder states in isolation that we know of: in our panel, T5 encoders collapse to or near chance at mid-depth. A 2×2 control crosses T5 pretraining with an embedding objective [[Pending: P2x2: raw encoder-only and embedding-trained T5 cells](#)]. (2) We compare anisotropy statistics with retrieval layer by layer. Top-PC share and effective rank separate collapsed from healthy layers, whereas mean pairwise cosine is lower at collapsed layers than at healthy ones. A train-only projection restores every

layer, which shows that the signal was present in the layer but masked. (3) We give a falsifiable test of the massive-coordinate account for mean-pooled passage retrieval: coordinate ablation, token audit, subspace split, and a D ablation [[Pending: E1–E3, H1](#)]. (4) We release the frozen testbed and split, the evaluation code, and lexical reference scores (Section 3).

2 Related Work

Anisotropy, outlier dimensions, and post-hoc correction. Contextual embeddings crowd into a narrow cone, usually measured by the mean cosine of random pairs ([Gao et al., 2019](#); [Ethayarajh, 2019](#)). That statistic mainly reflects a shared mean offset ([Rudman et al., 2022](#)). A separate distortion comes from a few outlier dimensions of very large magnitude ([Kovaleva et al., 2021](#); [Puccetti et al., 2022](#)), which dominate cosine similarity ([Timkey and van Schijndel, 2021](#)). In decoder language models they appear as massive activations that emerge early and persist through the middle layers ([Sun et al., 2024](#)). T5 encoders carry such outliers too, and their magnitude grows with depth ([Zhao et al., 2025](#)). Post-hoc corrections remove the mean and top principal directions ([Mu and Viswanath, 2018](#)), whiten the space ([Su et al., 2021](#); [Huang et al., 2021](#)), or down-weight frequent tokens ([Arora et al., 2017](#)). [Timkey and van Schijndel \(2021\)](#) apply such corrections at every layer of four pre-trained language models and recover word-similarity quality. After task fine-tuning, the dominant directions carry task information, and removing them hurts ([Rajaei and Pilehvar, 2021](#)). For final-layer embeddings, top-dimension variance share predicts when cosine fails ([Parupudi, 2026](#)), and effective rank predicts retrieval quality ([Rottach et al., 2025](#)). Other studies find that geometry statistics do not reliably predict task performance ([Nastase and Merlo, 2025](#); [Kulkarni et al., 2026](#)). None of these studies compares the statistics with retrieval quality layer by layer in T5 encoders. We make that comparison and fit the correction on training data only.

Layerwise similarity and retrieval quality. Intermediate layers often outperform the final layer on embedding tasks ([Skean et al., 2025](#)). Layerwise geometry studies place the anisotropy peak in the middle layers of decoder models ([Razzhigaev et al., 2024](#); [Machina and Mercer, 2024](#)) and report flatter profiles for encoders ([Razzhigaev et al.,](#)

2024). Hämmerl et al. (2023) find a large outlier dimension in a middle layer of a multilingual encoder and improve cross-lingual retrieval by zeroing it. Evidence for T5 stops short of its encoder layers. Mean-pooled final-layer T5 embeddings score poorly on similarity (Ni et al., 2022a). Whitening gives T5 large similarity gains, reported for an average of its first and last layers (Huang et al., 2021). Layerwise anisotropy for T5 has been reported with encoder and decoder states concatenated (Godey et al., 2024). We report the first per-layer retrieval profile of T5 encoder states in isolation that we know of.

Latin encoders and text reuse. Latin has dedicated encoder-only models, Latin BERT (Bamman and Burns, 2020) and LaBERTa and PhilBERTa, and the encoder-decoder models LaTa and PhilTa trained on the same data (Riemenschneider and Frank, 2023). Work on allusive text reuse in Latin finds that distributional semantics moderately improves retrieval over lexical matching (Manjavacas et al., 2019). Recent benchmarks target intertextuality in classical Latin literature (Schelb et al., 2026) and semantic search in Latin and Ancient Greek (Toyin et al., 2026). Reichbauer et al. (2026) mine parallel Latin and Greek sentences and find that whitening helps specialized encoders and slightly hurts LaBSE. Our testbed differs in its labels, which come from an edition of medieval canon-law witnesses, and in its question: where the retrieval signal sits inside each encoder.

3 Testbed and Protocol

Corpus. The Carolingian Canon Law project (CCL; Firey, 2009) is an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts that keys each transcribed unit to the source it copies. A *directory* is one source, and a *witness* is one manuscript’s copy of it; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display (Appendix A), so no model shaped them. The material is licensed CC BY-NC-SA 4.0 with attribution to the project. We will release the frozen testbed and split under the same license, in coordination with the project’s versioned exports, together with the evaluation code and lexical reference scores.

Split. We divide the witnesses into 847 training and 858 test files, and pairs form only within a split.

Singleton directories (545 of 840) are divided file by file, doubletons go whole to one side, and larger directories are divided internally so that every size class yields training positives. This gives 565 positive training and 596 positive test pairs; 535 test files have a same-directory partner in test (*existing*) and 323 do not (*new*).

Tasks and metrics. Task A scores every pair of test witnesses by the cosine of their pooled embeddings, and we report AUROC with same-directory pairs as positives. Task B is open-set routing: file an unidentified witness under a known directory, or flag it as new. Each test file scores every test directory by its maximum cosine to a member. A pseudo-directory NEW scores the threshold τ . The top-scoring entry is the decision. Directory accuracy at rank 1 (DirAcc@1) requires that entry to be the labeled directory, or NEW for a new file. Assignment accuracy scores only the existing-versus-new decision. Labeling every file as existing already scores 62.4%, so we report assignment accuracy alongside DirAcc@1.

Per layer we also read five statistics: three label-free geometry statistics and two label-based separation indices. The *mean pairwise cosine* over all pairs of pooled test passages is the usual anisotropy statistic, adapted from the word-level statistic of Ethayarajh (2019), which uses random pairs. Let $p_i = \lambda_i / \sum_j \lambda_j$ for the covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the centered training embeddings. The *top-PC share* (the share of centered variance on the first principal component) is p_1 . The *effective rank* is $\exp(-\sum_i p_i \log p_i)$, the entropy effective rank of Roy and Vetterli (2007) applied to the covariance eigenvalues; it is 1 when one direction holds all variance. The *cosine gap* g is the mean same-directory minus the mean different-directory test cosine, and the *standardized gap* g/s divides it by the standard deviation s of all pairwise test cosines, putting models with different cosine spreads on one scale. The two gaps need labels.

Train-only rule. A headline cell reads each model and setting at the layer with the highest training AUROC for Task A, or training DirAcc@1 for Task B (Table 9). The threshold τ maximizes training F1 over same- versus different-directory pairs. SIF frequencies are fit on training texts, and ABTT components and D on training embeddings. Layerwise profiles show every layer and select nothing. The rule selects every layer, threshold, and projection from training data alone. Analysis quantities

(the collapse cutoff and the worst layers) use test AUROC and select no reported headline cell. Appendix I uses a predeclared variant of the rule for attribution layers.

Post-processing. We mean-pool the hidden states of every transformer block, dropping only tokenizer artifacts without lexical content. All-but-the-Top (ABTT; Mu and Viswanath, 2018) subtracts the training mean and removes the top D principal components of the centered training embeddings from all vectors, with $D \in \{1, 2, 3, 5, 7, 10\}$ chosen per layer by training DirAcc@1. We write ABTT-only when we contrast it with SIF+ABTT. SIF (Arora et al., 2017) instead reweights each token by $a/(a + p(w))$ before pooling ($a = 10^{-3}$), with $p(w)$ estimated on training texts, and gives special tokens, such as the end-of-sequence token, zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both. Full-rank PCA whitening (Su et al., 2021) is ill-conditioned here, since it estimates 768 to 1,024 dimensions from 847 training vectors. It degenerates the learned threshold and also ranks below ABTT at every layer, at Task A AUROC 0.82–0.88 for the T5 encoders and LaBSE and 0.55–0.64 for the two decoders, so we exclude it (Appendix G). Whitening to a reduced dimension, as Su et al. (2021) also do, is the fairer comparison: [Pending: WHITEN: PCA whitening with $k \in \{64, 128, 256\}$, Task A AUROC].

Model panel. Table 1 crosses two factors. The first is the architecture with its pretraining objective: a T5 encoder with span-corruption pretraining, or an encoder-only model with masked-language-model pretraining. The second is whether a contrastive embedding objective follows pretraining. The T5 encoders we have run are the Latin LaTa, its trilingual sibling PhilTa (Riemenschneider and Frank, 2023), and mT5-base (Xue et al., 2021). The controls we have run are LaBSE (Feng et al., 2022), Qwen3-0.6B (Zhang et al., 2025), and KaLM-mini (Hu et al., 2025). In the models run so far the two factors are confounded: every T5 encoder is raw and every control is embedding-trained, so a T5-only collapse could reflect either. The pending rows break the confound. LaBERTa and PhilBERTa share authors and pretraining data with LaTa and PhilTa (Riemenschneider and Frank, 2023), and Latin BERT (Bamman and Burns, 2020) is a second raw Latin encoder. Sentence-T5 (Ni et al., 2022a) fills the embedding-trained T5 cell, and T5-

Model	Arch.	Data	Emb. obj.	L	Status
LaTa	T5	Latin	none	12	run
PhilTa	T5	Grc/La/En	none	12	run
mT5-base	T5	101 lang.	none	12	run
T5-v1.1-base	T5	English	none	12	[P2x2]
LaBERTa	Enc.	Latin	none	12	[P2x2]
PhilBERTa	Enc.	Grc/La/En	none	12	[P2x2]
Latin BERT	Enc.	Latin	none	12	[P2x2]
Sentence-T5 (or GTR-T5)	T5	English	contrastive	12	[P2x2]
LaBSE	Enc.	109 lang.	contrastive	12	run
Qwen3-0.6B	Dec.	multiling.	contrastive	28	run
KaLM-mini	Dec.	multiling.	contrastive	24	run

Table 1: The model panel as a 2×2 design. Arch.: a T5 encoder with span-corruption pretraining, or an encoder-only model (Enc.) with masked-language-model pretraining; LaBERTa and PhilBERTa are RoBERTa-style. Emb. obj.: the embedding objective trained after pretraining, if any. Blocks: raw T5 encoders; raw encoder-only models; embedding-trained T5 and encoder-only models; embedding-trained decoders (Dec.) outside the design. T5-v1.1-base is the matched raw partner of Sentence-T5. Data: pretraining languages (Grc/La/En: Ancient Greek, Latin, English). L: transformer blocks read (encoder blocks for T5). KaLM-mini runs its decoder with bidirectional attention. [P2x2]: not yet run.

v1.1-base (Raffel et al., 2020) is its matched raw partner.

Lexical reference. Character 3–5-gram TF-IDF, fit on training files, reaches 0.987 AUROC on Task A, tying the best pre-trained encoder cell, and 89.9 DirAcc@1 on Task B, above every pre-trained one. In every tercile of n-gram overlap among positive test pairs (terciles cut on the TF-IDF score itself), it ranks a witness’s partner first at least as often as every baseline and ABTT encoder configuration at its train-selected layer. The equivalence signal is therefore present in the surface text. Our claims concern where the retrieval signal sits inside encoders and what masks it, not whether embeddings beat n-grams on this corpus. Table 22 (Appendix H) gives all three lexical scorers.

4 Two Geometries

This section asks whether the anisotropy statistic that practitioners read tracks retrieval layer by layer. We first give the retrieval profile of every model, then set the geometry statistics of Section 3 against it.

Retrieval by depth. Figure 1 plots test AUROC against depth for all six models. The three T5 encoders fall to or near chance in their middle layers.

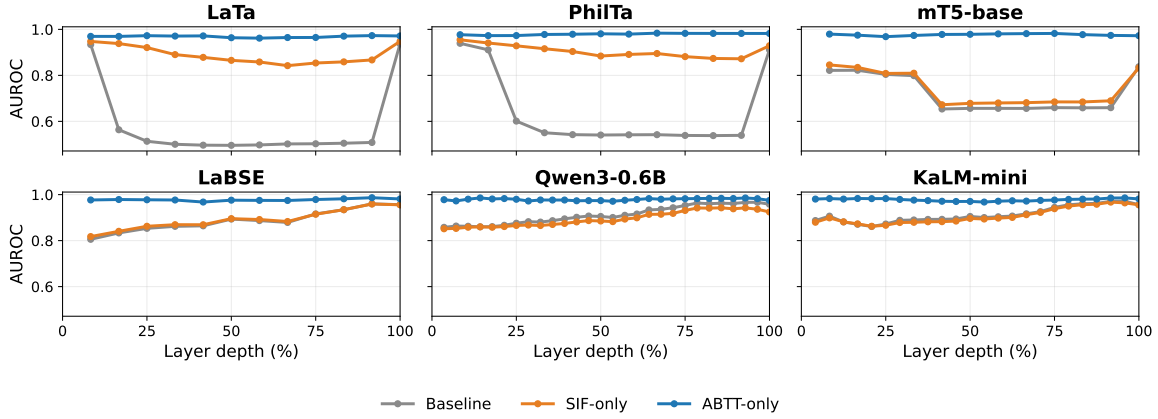


Figure 1: Pairwise retrieval AUROC (Task A, test split) of the cosine between mean-pooled passage vectors, against normalized layer depth (layer ℓ of L is plotted at $100\ell/L$ percent). Top row: the three T5 encoders. Bottom row: the three embedding-trained models. Gray: baseline mean pooling. Orange: SIF-weighted pooling. Blue: ABTT, with the number of removed components D chosen per layer on training data. AUROC 0.5 is chance. The T5 baselines fall to or near chance in their middle layers (LaTa 0.496 at layer 6, PhilTa 0.538 at layer 10, mT5-base 0.654 at layer 5) and recover at the last layer. The embedding-trained baselines stay above 0.80 and rise broadly with depth. ABTT holds every layer of every model between 0.96 and 0.99. SIF lifts LaTa and PhilTa and barely moves mT5-base.

Baseline AUROC bottoms out at 0.496 for LaTa at layer 6, at 0.538 for PhilTa at layer 10, and at 0.654 for mT5-base at layer 5. We call a layer *collapsed* when its baseline AUROC is below 0.70. By this definition, 26 of the 36 T5 layers collapse: layers 2–11 of LaTa, 3–11 of PhilTa, and 5–11 of mT5-base. We set this cutoff after inspecting test AUROC; it describes the profiles and selects no headline cell. Each T5 encoder recovers at its last layer, where LaTa reaches 0.938. The three embedding-trained models never fall below 0.806, and they improve broadly with depth to a best layer of 0.960 to 0.972. In our panel the collapse is thus confined to the T5 encoders, which are also the only models without an embedding objective (Section 6). ABTT, fit on training embeddings, restores every layer of every model (Section 7). The question is whether a geometry statistic would have flagged the collapsed layers.

Mean pairwise cosine. The usual diagnostic gives no warning of the collapse. Table 2 summarizes each model’s geometry over its layers. Mean pairwise cosine, averaged over all pairs of test passages, is highest in the embedding-trained models. Their mean cosine ranges from 0.585 to 0.974 across layers, while their retrieval stays above 0.80. At the collapsed T5 layers the statistic ranges from 0.23 to 0.62. LaTa (0.23–0.30) and mT5-base (0.30–0.36) sit below every embedding-trained layer, and PhilTa (0.53–0.62) overlaps the bottom of that range. Within the T5 encoders the statistic runs against the usual reading. Collapsed

layers have lower mean cosine than healthy ones, so it correlates with AUROC at Spearman +0.67 (Table 3). Within the embedding-trained models it carries almost no information (Spearman +0.10, Pearson −0.06). Computing it on training passages instead gives the same picture (Spearman +0.72 within T5, +0.12 within the embedding-trained models). A practitioner who reads a high mean cosine as a warning would flag the healthy models and pass the collapsed layers.

Variance statistics. The variance statistics separate the two groups, with one exception (mT5-base layer 4). We compute top-PC share and effective rank on centered training embeddings, so the shared offset that drives mean cosine does not enter them. At the collapsed T5 layers the top principal component holds 76 to 100 percent of the variance, and effective rank lies between 1.0 and 4.6. The two transition layers at the onset of collapse, LaTa layer 2 and PhilTa layer 3, set the lower end of the share and the upper end of the rank. Without them the ranges narrow to 83–100 percent and 1.00–2.17. At mT5-base layers 5–11 one direction holds all the variance to three decimals. In the embedding-trained models the largest top-PC share at any layer is 0.18 for Qwen3-0.6B, 0.42 for KaLM-mini, and 0.52 for LaBSE. Their effective rank ranges from 16 to 150. ABTT with $D = 10$ removes the concentration. On the training vectors it is fit on, this follows by construction. On test passages at the collapsed layers, top-PC share falls to 0.03–0.05 and effective rank rises to 148–173.

Model	Type	L	Geometry (raw view)				Baseline test AUROC		
			Mean cos.	Cos. SD	Max PC1	Min rank	ℓ^*	at ℓ^*	Min (layer)
LaTa	T5 raw	12	0.23–0.58	0.10–0.63	0.952	1.36	4	0.500	0.496 (6)
PhilTa	T5 raw	12	0.53–0.71	0.09–0.46	0.858	1.81	6	0.541	0.538 (10)
mT5-base	T5 raw	12	0.30–0.95	0.05–0.76	1.000	1.00	5	0.654	0.654 (5)
LaBSE	BERT emb.	12	0.73–0.93	0.04–0.10	0.524	16.19	8	0.879	0.806 (1)
Qwen3-0.6B	Decoder emb.	28	0.85–0.97	0.01–0.04	0.183	71.60	16	0.910	0.858 (1)
KaLM-mini	Decoder emb.	24	0.58–0.93	0.02–0.07	0.419	23.83	5	0.861	0.861 (5)

Table 2: Geometry and retrieval regimes of the six encoders in the raw view (mean-pooled hidden states with no post-hoc correction). Type: T5 raw is a T5 encoder pretrained without a sentence-embedding objective; emb. marks a BERT encoder or a decoder trained with one. L : number of transformer layers. Mean cos. and Cos. SD: range over layers of the mean and the standard deviation of the cosine over all pairs of test passages. Max PC1: peak top-PC share over layers, the largest share of variance on the first principal component of the centered train embeddings. Min rank: smallest entropy effective rank of the same centered train embeddings over layers. All four geometry columns are label-free. ℓ^* : layer where the PC1 share peaks. The last two columns give the baseline test pairwise AUROC of cosine at ℓ^* and its minimum over layers, with the layer in parentheses. Layer 1 is the first transformer block.

Statistic	All (100)		T5 (36)		Non-T5 (64)		LOMO ρ	
	r	ρ	r	ρ	r	ρ	min	max
<i>Target: baseline AUROC</i>								
Mean pairwise cosine	0.83	0.55	0.61	0.67	−0.06	0.10	0.42	0.67
Top-PC share	−0.92	−0.77	−0.86	−0.50	−0.47	−0.43	−0.83	−0.70
Effective rank	0.80	0.77	0.87	0.53	0.48	0.40	0.70	0.85
Raw cosine gap	0.24	0.22	0.63	0.72	0.15	−0.03	−0.10	0.48
Standardized gap	0.86	0.97	0.87	0.96	0.89	0.93	0.96	0.98
<i>Target: ABTT AUROC gain</i>								
Mean pairwise cosine	−0.83	−0.54	−0.60	−0.66	0.07	−0.07	−0.66	−0.41
Top-PC share	0.92	0.76	0.86	0.50	0.45	0.41	0.69	0.83
Effective rank	−0.80	−0.76	−0.88	−0.52	−0.44	−0.37	−0.84	−0.68
Raw cosine gap	−0.23	−0.23	−0.61	−0.71	−0.16	0.00	−0.48	0.09
Standardized gap	−0.85	−0.97	−0.87	−0.95	−0.88	−0.92	−0.98	−0.96

Table 3: Correlation between per-layer geometry and retrieval, one observation per model-layer, raw view. r : Pearson; ρ : Spearman. Columns pool all 100 model-layers, the 36 layers of the three T5 encoders, or the 64 layers of the three embedding-trained models; LOMO gives the minimum and maximum Spearman over the six leave-one-model-out subsets. Mean pairwise cosine is on test passages; top-PC share and effective rank are on centered train embeddings. These three statistics are label-free. The two gaps are label-based separation indices. Raw cosine gap is the mean test cosine of same-directory pairs minus that of different-directory pairs; standardized gap divides it by the standard deviation of all test pairwise cosines at that layer. Targets are the baseline test AUROC and the ABTT gain (AUROC with ABTT at its train-selected D minus baseline AUROC). The standardized gap is a signal-to-noise ratio of the kind AUROC summarizes, so it tracks AUROC by construction. ABTT AUROC varies little (0.962–0.987), so the gain correlations mirror the baseline ones. The pooled Pearson and Spearman values are both inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent.

What the statistics predict. Figure 2 sets both kinds of statistic against retrieval over all 100 model-layers, and Table 3 gives the correlations. Across the panel, top-PC share correlates with baseline AUROC at Spearman -0.77 and effective rank at $+0.77$, against $+0.55$ for mean pairwise cosine. Both coefficients are inflated by the two clusters of collapsed and healthy layers visible in panel (a), and the Pearson value of -0.92 most of all. Layers within a model are also not independent, so the 100 model-layers carry the evidence of roughly six clusters. Leaving out any one model keeps

the top-PC Spearman between -0.70 and -0.83 ; this shows robustness to single models, not to the clustering. A threshold states the separation more directly. A top-PC share of 0.6 or more marks all 26 collapsed layers with one false alarm, mT5-base layer 4, and every cutoff from 0.53 to 0.76 gives the same split. Like the collapse cutoff, this threshold is descriptive and was read off the data. Within a family, the variance statistics track AUROC only weakly. Top-PC share gives Spearman -0.50 within the T5 encoders and -0.43 within the embedding-trained models; effective rank gives

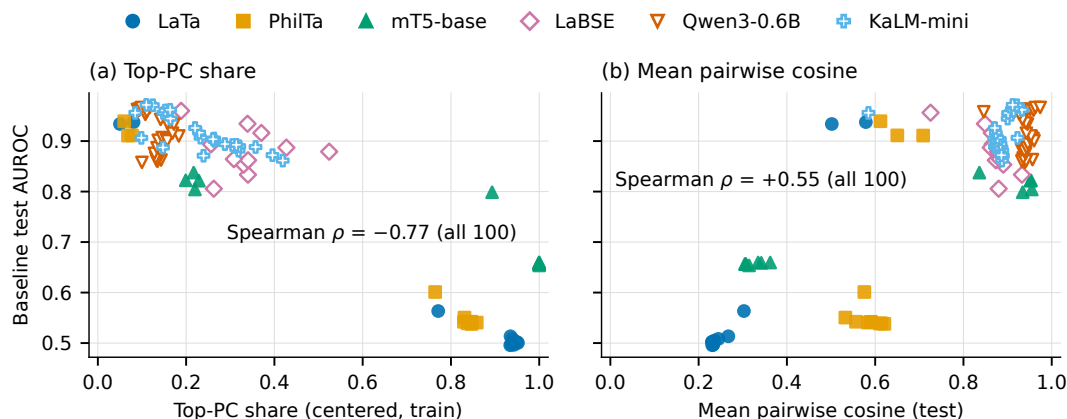


Figure 2: Per-layer geometry against baseline retrieval, one marker per model-layer (100 in all). Filled markers: T5 encoders (LaTa, PhilTa, mT5-base). Hollow markers: embedding-trained models (LaBSE, Qwen3-0.6B, KaLM-mini). Vertical axis: baseline Task A test AUROC. (a) Top-PC share, the share of centered training variance on the first principal component. (b) Mean pairwise cosine over all pairs of test passages, the usual anisotropy statistic. In (a) every collapsed layer (AUROC below 0.70) has a share of at least 0.76, and every other layer except mT5-base layer 4 (share 0.89, AUROC 0.80) lies below 0.53. In (b) the collapsed LaTa and mT5-base layers sit below every embedding-trained layer, and the collapsed PhilTa layers overlap the bottom of that range, so a high mean cosine gives no warning of collapse.

+0.53 and +0.40. Within single models, the top-PC Spearman runs from -0.28 (LaBSE) to -0.94 (mT5-base) over 12 to 28 layers. mT5-base layer 4 shows the limit. Its top-PC share is 0.893 and its effective rank 2.15, yet its AUROC of 0.799 is close to that of its healthy layers 1–3. The lower block of Table 3 adds no separate evidence. ABTT lifts every layer into a narrow band (0.962 to 0.987), so its gain is nearly a constant minus the baseline, and the gain correlations mirror the baseline ones. The variance statistics thus separate the collapsed regime from the healthy one. They do not rank layers within a regime, and a high share alone does not guarantee collapse. This limit fits reports that geometry statistics predict task performance unreliably (Nastase and Merlo, 2025; Kulkarni et al., 2026; Machina and Mercer, 2024).

The standardized gap. The raw cosine gap, the margin a threshold works with, tracks retrieval even less well. Across the panel it correlates with AUROC at Spearman $+0.22$. The standardized gap reaches $+0.97$, and for the AUROC target every Spearman column of Table 3 keeps it between $+0.93$ and $+0.98$. This agreement is expected by construction. The standardized gap is a label-based signal-to-noise ratio, close to the d' from which a binormal model computes AUROC, so we do not count it as a predictor. Its use here is to dissolve the mT5-base paradox. mT5-base shows why the two gaps differ (Figure 5, Appendix B). Its raw gap

rises from about 0.03 to 0.21–0.25 exactly where ranking fails, and within the model it correlates with AUROC at -0.78 . Over the same layers, the standard deviation of all pairwise cosines grows from 0.05–0.07 at layers 1–3 to about 0.75. Divided by that spread, the gap falls to 0.28–0.33, below the 0.41–0.49 of layers 1–3 and the 0.78 of layer 12. The within-model correlation then turns to $+0.52$. A gap that rises as ranking fails is thus no paradox: the collapsed layers spread all cosines widely, and same-directory pairs gain little relative to that spread. The resolution is partial. At mT5-base layer 4 the standardized gap is the lowest of the model (0.19), yet AUROC is 0.80. LaTa adds an anomaly that the standardized gap does not explain. Its raw gap is negative at layers 3–11, between -0.05 and -0.07 , so same-directory pairs score slightly below different-directory pairs on average. We do not have an account of this yet, and the token audit of Section 5 tests one candidate (E2).

SIF at the collapsed layers. SIF offers a first hint about what feeds the dominant direction. It down-weights frequent tokens and drops special tokens before pooling (Section 3), and its effect at the collapsed layers splits the T5 encoders. It lifts LaTa from 0.50–0.56 to 0.84–0.94 and PhilTa from 0.54–0.60 to 0.87–0.93. It moves mT5-base only from 0.65–0.66 to 0.67–0.69. ABTT reaches 0.96–0.98 at the same layers in all three. This

pattern suggests that tokens SIF down-weights or drops carry the dominant direction in LaTa and PhilTa, and that tokens it keeps carry it in mT5-base. Because SIF differs from mean pooling in two ways, the token audit of Section 5 separates them (E2).

Two regimes. The panel therefore shows two regimes. In the embedding-trained models every passage shares a large mean vector, which raises every cosine, and variance spreads over many directions. These models rank well but still gain from the projection. Their baseline AUROC is 0.806 to 0.972, and ABTT adds 0.007 to 0.031 at their respective train-selected layers (Section 7). In the middle layers of the T5 encoders one direction takes most of the variance, and cosine ranking falls to or near chance. The cosines there are spread across both signs rather than moderate. At mT5-base’s collapsed layers the 5th percentile of test cosines lies between -0.83 and -0.77 , the median between 0.84 and 0.90 , and the standard deviation about 0.75 . At LaTa’s collapsed layers the 5th percentile lies between -0.50 and -0.26 , and the standard deviation reaches 0.63 . Prior work already separates these two components. Timkey and van Schijndel (2021) distinguish a constant offset from a few high-variance dimensions that dominate cosine, and IsoScore (Rudman et al., 2022) shows that the mean cosine of random pairs mainly measures the offset. For final-layer embeddings, top-dimension variance share predicts cosine failure better than mean cosine (Parupudi, 2026), and effective rank predicts retrieval quality across many models (Rottach et al., 2025). Our contribution is the layerwise version of this comparison within one testbed, including raw T5 encoder states, where the low-rank regime coincides with chance retrieval. Section 5 asks what produces that dominant direction.

5 Localizing the Collapse

Section 4 showed that collapsed T5 layers concentrate their centered variance on one direction. Their cosines also take both signs, so the pooled vectors lie on both sides of one axis. The projection on the dominant direction therefore changes sign across passages, which a shift shared by all passages cannot produce. Prior work on outlier dimensions offers a candidate account. In pre-trained transformers a few residual coordinates reach magnitudes far above the median (Kovaleva et al., 2021;

Puccetti et al., 2022) and dominate cosine similarity (Timkey and van Schijndel, 2021). In decoder language models these massive activations take near-constant values on the tokens that carry them (Sun et al., 2024), and T5 encoders carry such outliers too (Zhao et al., 2025). Their prevalence tracks token frequency in pretraining (Puccetti et al., 2022). Mean pooling adds one step. It averages the per-token values over a token mix that differs from passage to passage, so the pooled coordinate varies across passages. If that variation is large, cosine ranks passages by one coordinate and ignores their content.

Preliminary evidence fits this account. At LaTa layer 7, the layer the attribution rule selects (Appendix I), three coordinates have a mean magnitude of about 5,000 against a median of 36. Their share of the pooled cosine, measured as the summed per-coordinate contribution of Timkey and van Schijndel (2021), is [E1: LaTa layer 7 share, to recompute]. Massive coordinates alone do not collapse retrieval, however. At mT5-base layer 1, three coordinates carry [E1: mT5-base layer 1 share, to recompute] of the cosine, yet AUROC is 0.82 and top-PC share is 0.23. The refined hypothesis is that collapse occurs when the pooled massive coordinates vary across passages at least as much as they shift all passages together. We make this measurable with r , the standard deviation of a pooled coordinate across training passages divided by the absolute value of its mean. The hypothesis predicts $r \geq 1$ for the top coordinates at collapsed T5 layers, and $r < 1$ at mT5-base layer 1 and in Qwen3-0.6B. A high centered top-PC share already means that one direction varies across passages. The testable content is that this direction is aligned with a few coordinates and has token carriers. The four tests below each state a prediction and its alternative. Every intervention is fit on training embeddings only.

Coordinate ablation (E1). Zeroing a single outlier dimension already improves cross-lingual retrieval in a multilingual encoder (Hämmerl et al., 2023). We test whether such coordinates explain a collapse to chance, and at which depths. At every layer of LaTa, PhilTa, mT5-base, LaBSE and Qwen3-0.6B, we rank residual coordinates two ways on training passages: by mean absolute value, and by variance across passages. For each ranking we zero the top k coordinates, $k \in \{1, 3, 5, 10\}$. We also standardize every coordinate by its train-

ing mean and standard deviation (Timkey and van Schijndel, 2021), and we center the vectors without removing any component ($D = 0$). ABTT with $D = 1$ serves as a reference, since it repairs a rank-one nuisance under either account. If a few passage-varying coordinates cause the collapse, zeroing the top $k \leq 5$ coordinates by variance restores AUROC at the collapsed T5 layers to at least 0.90. Zeroing by mean magnitude, which also picks coordinates that shift all passages together, should help less. Top-PC share then falls toward the range of the non-T5 models. For LaBSE and Qwen3-0.6B, which have no collapsed layer, zeroing should change AUROC by at most 0.03 at their train-selected layers, where ABTT itself adds 0.007 to 0.031, and should recover only a small part of the ABTT gain at their weakest layers. If standardization repairs a collapsed layer but zeroing $k \leq 5$ does not, the nuisance sits on a few coordinates, but more than five. If neither repairs it and only a projection does, the dominant axis is spread over many coordinates of similar variance. The account then reduces to “one dominant direction”, which is what Section 4 already reports. Centering alone tests whether the shared offset matters at all at these layers. Table 4 gives each model at its worst baseline layer. We find [Pending: E1: AUROC and top-PC share after zeroing the top- k coordinates under each ranking, after standardization, and after centering, per layer; expected: zeroing $k \leq 5$ coordinates by across-passage variance restores collapsed T5 layers to AUROC ≥ 0.90 and top-PC share < 0.2 , and LaBSE and Qwen3-0.6B change by at most 0.03 at their train-selected layers].

Token audit (E2). The account names a carrier: some tokens hold the massive values. For each collapsed T5 layer we record which tokens carry the top coordinates, their sign and magnitude, and the ratio r of the pooled coordinates. We also relate each passage’s score on the top component to its log length and to its share of the 100 most frequent tokens. SIF sharpens the test. It rescues LaTa and PhilTa but not mT5-base (Section 4), and it differs from mean pooling in two ways: it down-weights frequent tokens, and it drops special tokens such as the end-of-sequence token. A control separates the two: mean pooling with special tokens dropped and no frequency weighting. If that control alone rescues LaTa and PhilTa, the end-of-sequence token carries their coordinates; if only full SIF does, frequent tokens do. For mT5-base the evidence

points away from the end-of-sequence token. If that token carried its coordinates, SIF would delete it and rescue mT5-base. Since SIF does not, mT5-base’s carriers must be tokens that SIF keeps. If frequent tokens carried the coordinates in all three models, the SIF split would need another explanation, and the frequency link would weaken. The same audit on Qwen3-0.6B tests the refined hypothesis. Massive coordinates there should shift all passages together ($r < 1$). If they instead vary across passages as much as in the T5s, the refinement is false. The audit also addresses LaTa’s negative raw gap: same-directory pairs at its middle layers score slightly lower than different-directory pairs. If the top component tracks passage length, and same-directory witnesses differ in length more than different-directory pairs do, this gap follows. Comparing the absolute difference in log length between the two kinds of pair checks the second condition directly. We find [Pending: E2: carrier tokens per model and layer; ratio r of the pooled coordinates; AUROC of mean pooling without special tokens; correlation of the top-component score with log length and frequent-token share; expected: frequent tokens carry the coordinates in LaTa and PhilTa, tokens that SIF keeps carry them in mT5-base, and Qwen3-0.6B coordinates have $r < 1$].

Subspace split (E3). Our account says that the top D directions ABTT removes hold nuisance in pre-trained spaces. We rank passages by their projection onto PC1 alone, onto PCs 2 to D , and onto the retained subspace, and we compare the removed with the retained subspace. At collapsed layers the full vector already ranks at chance and the removed subspace dominates it, so the test informs mainly where the full vector is healthy. The cleanest contrast is LaTa’s layer 12, which the pre-trained and fine-tuned models share. ABTT raises the pre-trained layer from 0.938 to 0.971 and lowers the fine-tuned one from 0.984 to 0.970. If the removed subspace holds nuisance in pre-trained spaces, it ranks well below the retained subspace there, although PCs 2 to D may still carry some topical signal. Fine-tuning is reported to concentrate task signal in the dominant directions (Rajaei and Pilehvar, 2021). If that holds here, the removed subspace of fine-tuned LaTa, Qwen3-0.6B and KaLM-mini ranks well above chance and close to the retained one. This would explain why ABTT lowers fine-tuned LaTa’s AUROC. If the removed subspace ranks no better after fine-tuning than before,

Model	L	Base	Zeroed, ranked by mean $ x $				Zeroed, ranked by variance				Std.	$D=0$	ABTT $_{D=1}$
			1	3	5	10	1	3	5	10			
LaTa	6	0.496	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]
PhilTa	10	0.538	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]
mT5-base	5	0.654	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]
LaBSE	1	0.806	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]
Qwen3-0.6B	1	0.858	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]	[E1]

Table 4: Coordinate ablation at each model’s worst baseline layer (L). Cells give Task A test AUROC for the unmodified mean-pooled vectors (Base); after zeroing the k residual coordinates with the largest mean absolute training value, or with the largest variance across training passages; after per-coordinate standardization with training statistics (Std.); after centering alone ($D=0$); and after ABTT with one component (ABTT $_{D=1}$), a reference that repairs a rank-one nuisance under either account. All statistics are fit on training embeddings only.

that drop needs another cause. We find [Pending: E3: Task A AUROC of PC1 alone, PCs 2 to D , and the retained subspace, pre-trained vs fine-tuned, at each model’s reported layer and at LaTa layer 12; expected: the removed subspace ranks well below the retained one in pre-trained models and close to it in fine-tuned LaTa].

How many directions (H1). ABTT selects $D = 10$, the top of the searched grid, in 87 of 100 model-layers (81 of 100 when SIF precedes it). Selection by training DirAcc@1 never picks $D \leq 2$: it picks $D = 3$ once, $D = 5$ twice, and $D = 7$ ten times. This already sits uneasily with a single nuisance axis, at least for routing. We therefore sweep $D \in \{0, 1, 2, 3, 5, 7, 10, 15, 20, 30, 50\}$ at all 100 model-layers, where $D = 0$ is centering alone, and read Task A AUROC. If the nuisance is rank-1, $D = 1$ recovers most of the AUROC gain at the collapsed T5 layers. We set the threshold at 80 percent of the $D = 10$ gain. The curve then stays flat beyond $D = 1$, and centering alone recovers little. If $D = 1$ falls below that threshold, the nuisance is a low-dimensional subspace, and we drop the one-direction wording. Figure 3 shows the sweep. We find [Pending: H1: AUROC and training DirAcc@1 against D for every model-layer; expected: $D = 1$ recovers ≥ 80 percent of the $D = 10$ AUROC gain at collapsed T5 layers and $D = 0$ recovers little].

Scope. We do not ask where these coordinates come from in pretraining, or why they peak at mid-depth in the T5 encoders of our panel. We ask only whether they explain the retrieval failure. Three outcomes would falsify the account. First, neither zeroing $k \leq 5$ coordinates nor standardization repairs the collapsed layers while ABTT does, which makes the nuisance a distributed direction. Second, Qwen3-0.6B shows passage-varying massive coordinates ($r \geq 1$) of T5 scale without collapse. Third,

[Pending: H1: figure not yet generated]

Figure 3: Task A test AUROC against the number of removed components D , with $D = 0$ meaning centering alone. One line per model at its worst baseline layer; ABTT is fit on training embeddings only. [Pending: H1: caption to be checked against the generated figure]

the removed subspace ranks as well as the retained one in the pre-trained T5s, which means ABTT discards signal along with nuisance. Under any of these, the paper’s claim reduces to the geometric description of Section 4 and the repair. Section 6 turns from the coordinates to a second question: whether T5 pretraining or the missing embedding objective produces them.

6 Objective or Architecture?

Our panel confounds two factors. Every T5 we have run is a raw pre-trained encoder. Every model that does not collapse is embedding-trained: LaBSE, Qwen3-0.6B and KaLM-mini. The collapse may follow T5 pretraining, or it may follow the absence of an embedding objective. The panel so far cannot separate the two, so the claim that collapse is T5-specific holds only in our panel.

Table 1 gives the design. The LaTa–LaBERTa and PhilTa–PhilBERTa pairs hold data and authors fixed and vary the architecture factor. The embedding-trained T5 cell holds Sentence-T5 (or GTR-T5) (Ni et al., 2022a,b). Its matched raw partner is T5-v1.1-base on the Latin corpus, since Sentence-T5 starts from an English T5 checkpoint rather than from LaTa.

The raw encoder-only cell is the discriminat-

Cell	Models	AUROC _{min}	PC1 _{max}
T5, raw	LaTa, PhilTa, mT5-base	0.50–0.65	0.86–1.00
T5, raw	T5-v1.1-base	[P2x2]	[P2x2]
T5, raw+FT	LaTa (fine-tuned)	0.50–0.57	[FT]
T5, emb.	Sentence-T5	[P2x2]	[P2x2]
Enc., raw	LaBERTa, PhilBERTa, Latin BERT	[P2x2]	[P2x2]
Enc., emb.	LaBSE	0.806	0.524

Table 5: Architecture-by-objective panel. AUROC_{min} is the lowest Task A test AUROC over layers of the unmodified mean-pooled vectors (a range when the cell holds several models; for fine-tuned LaTa, the range over layers 2–11). PC1_{max} is the peak top-PC share over layers, the largest share of centered training variance on the first principal component. Enc. is encoder-only, emb. is embedding-trained, and FT is contrastive fine-tuning on training pairs. T5-v1.1-base is read on the Latin corpus as the raw partner of Sentence-T5. [Pending: FT: import layerwise fine-tuned LaTa row]

ing test. Both accounts predict that the raw T5s collapse and that LaBSE does not. If LaBERTa and PhilBERTa keep retrieval at every depth, the collapse follows T5 pretraining. If they collapse, it follows the missing embedding objective. The embedding-trained T5 cell is weaker evidence, because its loss, like our fine-tuning below, attaches at the output. A Sentence-T5 that collapses at mid-depth is then predicted by both accounts, while one that stays healthy favors the objective account. We read this cell against LaBSE’s healthy middle layers and against raw T5-v1.1-base. An English model may also retrieve Latin poorly at every layer for reasons unrelated to collapse, so for this cell the label-free top-PC share and effective rank are the primary readout. Table 5 reports the lowest baseline AUROC over layers and the peak top-PC share of each cell. We find [Pending: P2x2: lowest baseline AUROC and peak top-PC share over layers for LaBERTa, PhilBERTa, Latin BERT, T5-v1.1-base and Sentence-T5; expected under the architecture account: the raw encoder-only models stay above 0.80 AUROC at every layer, while Sentence-T5 and T5-v1.1-base reach a top-PC share above 0.8 at mid-depth].

Fine-tuning gives a first piece of evidence. Contrastive fine-tuning on 499 of the training pairs (Appendix H) raises LaTa’s last layer from 0.938 to 0.984 AUROC (Table 21). Its layers 2 to 11 still collapse, at AUROC 0.50 to 0.57. Supervision attached to the output thus leaves the intermediate layers unrepaired. A similar output-only effect is reported for a multilingual encoder, where

sentence-level tuning removes an outlier dimension at the output but not at layer 8 (Hämmerl et al., 2023). This result does not separate the two accounts: small-scale supervision at the output may leave intermediate layers untouched under either. Our fine-tuning runs at most eight epochs on 499 pairs, far less than the large-scale contrastive training behind Sentence-T5, so the raw encoder-only cell remains the test.

All retrieval evidence comes from one labeled corpus. The geometry, however, needs no labels, so we can cross model with text. We compute top-PC share and effective rank per layer for mT5-base and PhilTa, whose pretraining covers English, on an English sample. We compute the same for T5-v1.1-base (Raffel et al., 2020) on both the Latin corpus and the English sample. If the collapse is a property of the model, each model shows the same profile on both texts: top-PC share above 0.8 and effective rank near 1 at mid-depth. If mT5-base stays high-rank on English, the collapse depends on the input text. If T5-v1.1-base stays high-rank on English while mT5-base collapses there, it depends on the pretraining data. We find [Pending: GEN: per-layer top-PC share and effective rank for mT5-base and PhilTa on the English sample, T5-v1.1-base on both texts (sample source and size to be fixed); expected: a mid-depth low-rank profile in every cell]. This check speaks to geometry only. Whether retrieval also collapses on English text stays outside what this testbed measures.

7 Repair and Its Limits

Sections 5 and 6 ask what masks the retrieval signal. This section asks whether a train-only projection can remove the mask, and where it stops helping. ABTT (Section 3) subtracts the training mean and removes the top D training components; no model parameter changes.

ABTT restores every layer. ABTT lifts every layer of every model to a pairwise AUROC between 0.96 and 0.99, including the T5 layers that sit at chance. At the train-selected layers of Table 6, baseline AUROC spans 0.838 to 0.972, and ABTT moves every model into a band from 0.971 to 0.987. The gain is largest where the baseline is weakest: mT5-base gains 0.137, while KaLM-mini gains 0.009. At each T5 encoder’s worst baseline layer, ABTT recovers AUROC to within 1.0 point of that model’s best ABTT layer. Because ABTT is a linear projection fit on training vectors, this shows

Model	Task A AUROC				Task A cosine gap			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	0.938	0.948	0.971	0.966	0.237	0.366	0.525	0.580
PhilTa	0.939	0.955	0.983	0.982	0.231	0.319	0.540	0.573
mT5-base	0.838	0.845	0.975	0.974	0.081	0.036	0.536	0.519
LaBSE	0.956	0.956	0.987	0.976	0.183	0.184	0.611	0.580
Qwen3-0.6B	0.966	0.942	0.973	0.972	0.024	0.053	0.553	0.540
KaLM-mini	0.972	0.964	0.981	0.981	0.056	0.056	0.568	0.535
LaTa (fine-tuned)	0.984	–	0.970	–	0.387	–	0.548	–
Qwen3-0.6B (fine-tuned)	0.996	–	0.994	–	0.684	–	0.716	–
KaLM-mini (fine-tuned)	0.997	–	0.994	–	0.636	–	0.651	–

Table 6: Task A pairwise duplicate detection for all six models under four post-processing settings. Each cell is a test-set score at the layer chosen by training-set AUROC, listed in Table 9; ABTT uses D tuned on train. Baseline AUROC spans 0.838 to 0.972, and ABTT lifts every model into a 0.971 to 0.987 band. Cosine gap is defined in Figure 4. Per-layer grids: Appendix Tables 10, 14, and 18. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev curve takes the rest), a ceiling at this training budget (setup: Appendix H). ABTT moves the fine-tuned encoders’ AUROC 0.984 to 0.970 for LaTa (gap 0.387 to 0.548), 0.996 to 0.994 for Qwen3-0.6B (gap 0.684 to 0.716), and 0.997 to 0.994 for KaLM-mini (gap 0.636 to 0.651).

that the retrieval signal was linearly present in every layer. The character 3–5-gram TF-IDF reference reaches 0.987 on the same split (Table 22). The repair therefore matches the lexical reference but does not exceed it.

Routing converges. Task B shows the same pattern at the directory level (Table 7). Under the baseline, DirAcc@1 spans 39.3 points across models, so model choice dominates. ABTT compresses that spread to 3.3 points, from 86.1 to 89.4. Assignment accuracy moves from a 40.3-point spread to a band from 88.5 to 91.7. Over five random reassignments of query and reference files, the baseline spread of 36.6 points falls to 1.7 under SIF+ABTT, with standard deviations of at most 1.0. The lexical reference routes at 89.9 on the single split, above every ABTT cell (86.1 to 89.4), and at 91.1 over five seeds, above every SIF+ABTT cell (88.9 to 90.6). Surface overlap remains the strongest router on this corpus. Under SIF+ABTT every model places the correct directory in a two-item shortlist for more than 95 percent of queries (Table 17, Appendix G). The single-split cells in both headline tables carry no intervals yet: **[Pending: CI: directory-level bootstrap intervals on every headline cell]**.

The offset regime and the open-set decision. The embedding-trained models do not collapse, yet ABTT still changes their routing. At their respective train-selected layers, their baseline AUROC was already 0.96 to 0.97. Under ABTT, assignment accuracy rises from 83.3 to 90.8 for LaBSE, from 82.3 to 91.5 for Qwen3-0.6B, and from 87.5 to 91.7 for KaLM-mini. ABTT also improves their rank-

ing: LaBSE’s AUROC rises from 0.956 to 0.987, and DirAcc@1 gains 3.5 to 9.1 points across the three. Whether centering alone, which removes the shared offset, accounts for this is what the $D = 0$ check below tests.

If an offset only added a constant to every cosine, order would be preserved and a threshold fit on training files would absorb it. A shared offset vector need not act as such a constant. Its effect on each cosine depends on the passage’s norm and its projection onto the offset, so it can reorder pairs. LaBSE’s AUROC gain under ABTT, which removes the mean and ten components, is consistent with this, and the $D = 0$ check isolates the mean. A cost to the open-set decision beyond ranking would need offsets that differ by query, such as hub vectors that sit close to many queries (Radovanović et al., 2010), or a shift between the training and test score distributions. The threshold grid offers a mundane alternative. We fit τ over 200 evenly spaced values in $[0, 1]$, a step of about 0.005, while the test cosines of Qwen3-0.6B spread with a standard deviation as small as 0.007. One grid step can then span most of a standard deviation, and ABTT could help routing simply by spreading the cosines. Five checks on cached embeddings test these readings: **[Pending: PQ: threshold-free existing-vs-new AUROC, oracle-vs-train threshold gap, hubness statistic, centering-only $D = 0$, and a threshold refit on a fine or quantile grid]**. Until they run, we do not attribute the routing gain to the offset geometry.

Model	Assignment accuracy				Directory accuracy @1			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	73.8	83.2	88.5	90.2	72.1	81.8	86.1	88.2
PhiTa	70.9	79.0	91.1	91.4	69.3	77.7	88.3	88.9
mT5-base	47.2	54.5	90.3	90.3	46.6	54.1	88.5	87.5
LaBSE	83.3	83.4	90.8	89.6	81.5	81.9	88.5	86.9
Qwen3-0.6B	82.3	76.0	91.5	90.8	80.3	74.2	89.4	88.6
KaLM-mini	87.5	85.7	91.7	90.8	85.9	84.0	89.4	88.0
LaTa (fine-tuned)	83.4	–	87.8	–	81.6	–	85.2	–
Qwen3-0.6B (fine-tuned)	91.4	–	92.1	–	90.8	–	91.4	–
KaLM-mini (fine-tuned)	92.4	–	93.4	–	91.7	–	92.5	–

Table 7: Task B autonomous routing for all six models under four post-processing settings, in percent. Each cell is a test-set score at the layer chosen by training-set directory accuracy at rank 1, listed in Table 9. Assignment accuracy scores the existing-versus-new decision alone, comparing each file’s maximum cosine against the train-fit threshold τ ; a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Directory accuracy at rank 1 requires the correct directory, so read the two columns together. Per-layer grids: Appendix Tables 11, 15, and 19. Below the rule, reference systems on the same split with the same evaluation code: LaTa, Qwen3-0.6B, and KaLM-mini fine-tuned contrastively on 499 of the 565 positive train pairs (a 28-directory dev carve takes the rest), a ceiling at this training budget (setup: Appendix H). The fine-tuned encoders with ABTT sit below every zero-shot ABTT cell for LaTa (87.8 and 85.2), above every zero-shot ABTT cell for Qwen3-0.6B (92.1 and 91.4), and above every zero-shot ABTT cell for KaLM-mini (93.4 and 92.5), against cells spanning 88.5 to 91.7 and 86.1 to 89.4.

SIF. SIF never matches ABTT on any model or task; we use it only as the diagnostic of Section 4 (full sweeps in Appendix G).

Where the repair costs. On pairwise ranking, ABTT lowers AUROC for every fine-tuned model. Fine-tuned LaTa falls from 0.984 to 0.970. Fine-tuned Qwen3-0.6B and KaLM-mini fall by 0.002 to 0.003, a margin we cannot separate from noise until the intervals land; for Qwen3-0.6B the two rows also sit at different train-selected layers. The same projection still raises their routing: DirAcc@1 rises from 81.6 to 85.2 for fine-tuned LaTa, from 90.8 to 91.4 for Qwen3-0.6B, and from 91.7 to 92.5 for KaLM-mini (Table 21). The ranking loss is consistent with Rajae and Pilehvar (2021), who found that removing top directions lowers the STS performance of BERT after Siamese fine-tuning. Reichbauer et al. (2026) find, by contrast, that whitening still helps fine-tuned Latin and Greek encoders. Why the projection costs ranking in fine-tuned models is the subspace question of Section 5: if fine-tuning moves task signal into the top components, then removing them removes signal (E3).

Fine-tuning on 499 training pairs also beats the training-free projection for the embedding-trained models. Over five random reassignments of query and reference files, fine-tuned Qwen3-0.6B with ABTT routes at 92.3 against 90.5 for its zero-shot ABTT row, and KaLM-mini at 93.0 against 90.9; for LaTa the two tie at 87.7 (Appendix H).

Practitioner rule. On this testbed, a top- D projection fit on training vectors helped every pre-trained encoder at every layer and hurt ranking in contrastively fine-tuned ones. A high top-PC share (0.76 or more here) flags a collapsed layer; mean pairwise cosine did not. Neither statistic chooses among healthy layers, and a character n-gram reference routes better than every pre-trained configuration on this corpus.

8 Discussion

The analysis describes one failure precisely. In the middle layers of the T5 encoders in our panel, one dominant direction [Pending: H1: or a low-dimensional subspace] takes over the pooled vectors, and cosine ranking falls to or near chance. Top-PC share separates these layers from healthy ones; mean pairwise cosine does not. [Pending: E1–E3: whether massive coordinates explain the direction]. A projection fit on training vectors removes the direction and restores ranking at every depth. Choosing among healthy layers still needs a train-only task signal.

Several observations remain open. Qwen3-0.6B and KaLM-mini are decoders of the kind in which massive activations are reported (Sun et al., 2024), yet they never collapse; the E2 audit tests whether their coordinates shift passages together instead of varying across them. At mT5-base layer 1, a few massive coordinates carry much of the pooled cosine, yet the layer still ranks at AUROC 0.82. In

LaTa’s middle layers the raw cosine gap is negative (-0.05 to -0.07), so same-directory pairs score slightly below different-directory pairs on average, while AUROC stays at chance. Section 4 shows that the standardized gap does not account for it, and the token audit (E2) tests a length account. Architecture and training objective are confounded in the models we have run, so the T5 attribution stays provisional until the controlled panel of Section 6 lands.

For the editor who needs to route new witnesses, the repair narrows the choice of model (Section 7). Under SIF+ABTT every model places the correct directory in a two-item shortlist for more than 95 percent of queries. The existing-versus-new call remains wrong for roughly one file in ten, so the open-set decision stays with the scholar. A character n-gram reference routes as well as any repaired pre-trained encoder on this corpus, and a practical tool should start from it. We release the frozen testbed and split, the evaluation code, and lexical reference scores so that others can test these accounts on their own encoders. Token attributions at the train-selected layers are reported in Appendix I; ABTT yields leave-one-out rank faithfulness gains there, with mixed ERASER-style rationale metrics.

Limitations

The study uses one labeled corpus in one language and one domain: medieval Latin canon law. The 1,705-witness corpus is large for this field but small by current NLP standards. We test six models, and others may behave differently. In the models we have run, architecture and training objective are confounded, so we cannot yet say whether the mid-depth collapse follows from T5 pretraining or from the absence of an embedding objective; the controlled panel of Section 6 is designed to separate them (P2x2). The headline cells carry no confidence intervals until the directory-level bootstrap is run (CI), so small differences, such as the ABTT drops on fine-tuned Qwen3-0.6B and KaLM-mini, should not be read as effects. ABTT needs enough text to estimate principal components, so smaller archives need separate study.

Task B measures ranking quality and a single threshold decision. It does not measure the full judgment a working scholar brings to deciding whether a witness belongs to an existing directory. The attribution analysis in the appendix is local to fixed query-candidate scores and failed its pre-registered threshold; its consistent effect is a leave-one-out rank faithfulness gain, and the ERASER-style rationale metrics are mixed.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project’s parallel display, not by independent double annotation, so no inter-annotator agreement figure exists for this testbed. Collation compares the witnesses of one key, so it has nothing to compare when a key has a single witness, and 545 of the 840 directories hold one witness in this testbed.

Ethical Considerations

The benchmark holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 licence (CCL, 2026a). The project has confirmed in writing that the licence covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project’s Advisory Board, attribution to the project suffices; the project’s interface credits the transcriber and proofreader of each unit. The project’s corpus keeps growing and being corrected, so benchmark

version 1 is a dated snapshot, fixed by a sha256 manifest of its 1,705 labelled files, and every number in this paper comes from it. We will release it under the same licence, in coordination with the project’s periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own.

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The project. The Carolingian Canon Law project (CCL; Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer (2014) reviews it.¹ Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript's copy of one unit (one transcribed canon or excerpt).

Transcription. Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative's XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks the text against the image to the letter and adds TEI markup (CCL, 2026b). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit (CCL, 2026a).

Identifiers. The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit's position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item's reference in modern scholarship, such as a canon,

letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

Labels. CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in CTOU.567.16; some labels carry a trailing annotation, as in CTOU.567.16 (15) or CARL.501?.18. Counting a directory as key-named when its label begins with a four-letter code, a period, a year and a period, whatever follows, 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (Can.apost.N), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

Assignment and verification. Keys were assigned by the project's director, a specialist in early medieval canon law, or by graduate students working under the director's supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project's parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project's texts, thousands of units still carry no key, and they are the queries that Task B is meant to serve.

Derivation. We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit's scholSource division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened (editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key

¹Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

receives no directory. The rule is scripted; checked against a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus, 100 percent after whitespace normalization with no character-level disagreement, and places every one of them in the same directory.

B Layer Diagnostics and Attribution Layer Selection

This appendix complements the layerwise view of Section 4. Figure 4 adds the raw cosine gap against depth for all six models to the AUROC profile of Figure 1; the gap rises in mT5-base’s middle layers while its ranking fails. Figure 5 plots the raw gap, the standardized gap, and baseline AUROC side by side. It also separates the most anisotropic layers, where geometry collapses, from the operational attribution layers, which are selected by training-set retrieval accuracy. Table 8 lists both for the three T5 encoders, together with the top-PC share and effective rank before and after ABTT.

Top-PC share and effective rank correlate with the ABTT gain across all 100 model-layers (Spearman +0.76 and −0.76, Table 3), but only because ABTT lifts every layer into a narrow band, so the gain mirrors the baseline. Within a family the correlations are weak. The statistics locate where ABTT helps; they do not select layers, which the train-only rule does.

C Score Distributions at the Collapsed Retrieval Layer

Figure 6 restates the collapse and the repair as score distributions rather than as summary metrics, for the model whose collapse is easiest to read.

D Expanded Per-Layer Results for T5 Encoders

The main text compresses the headline retrieval results into Tables 6 and 7, which report one train-selected layer per model and method; Table 9 lists those layers. This appendix provides the expanded per-layer Task A duplicate-detection, Task B autonomous-routing, and Task B top- k ranking tables for LaTa, PhilTa, and mT5-base; Appendix F gives the same view for the other three models.

E Last-Token vs. Mean Pooling

Decoder-only embedding models often use last-token pooling. We compare last-token pooling

with mean pooling under the same ABTT pipeline, train/test split, and ABTT choices. Table 13 reports, for each model and pooling, the layer chosen on the train split by DirAcc@1, the rule of Table 7, with test-set Task B assignment accuracy, DirAcc@1, and cosine gap at that layer. Mean pooling performs better on assignment accuracy for every model in this experiment, including the decoder-only models.

F Per-Layer Results for Non-T5 Models

The main text focuses on LaTa, PhilTa, and mT5-base. This appendix reports the same per-layer view for LaBSE, Qwen3-0.6B, and KaLM-mini under the same comparison: baseline mean pooling versus ABTT with D tuned per layer on the train split.

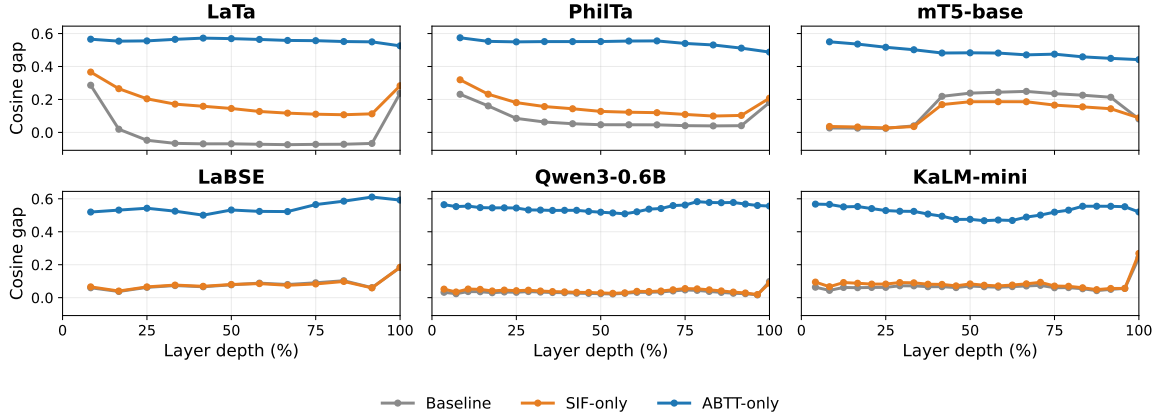


Figure 4: Raw cosine gap against normalized layer depth for all six models, test split, same panels as Figure 1. The gap is the mean cosine of same-directory pairs minus that of different-directory pairs, the margin a threshold works with. Gray: baseline mean pooling. Orange: SIF-weighted pooling. Blue: ABTT, with D chosen per layer on training data. Baseline gaps stay below 0.29 at every layer of every model. In the middle layers, LaTa’s gap turns slightly negative and PhilTa’s falls toward zero, while mT5-base’s gap rises as its ranking fails (compare Figure 1). Figure 5 resolves the mT5-base case by dividing the gap by the spread of all cosines. ABTT raises every model’s gap to between 0.44 and 0.61 at every depth. SIF raises the LaTa and PhilTa gaps and leaves the other four near or below baseline.

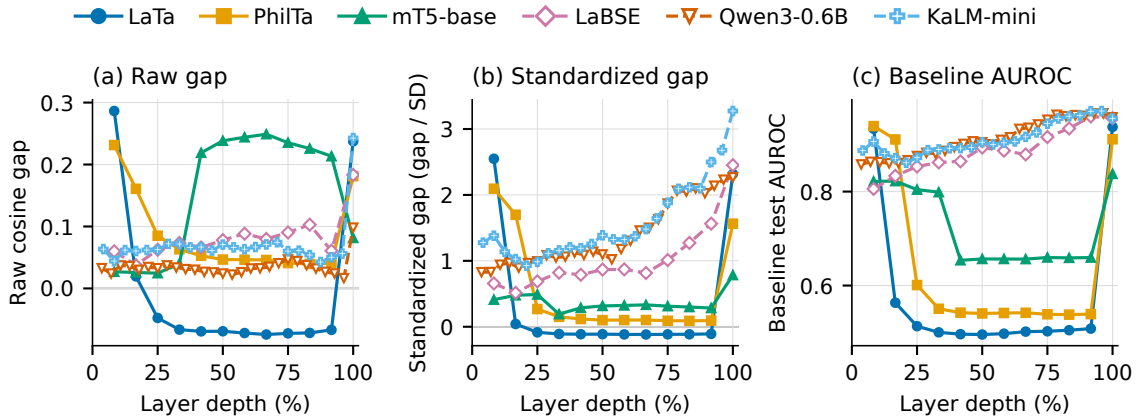


Figure 5: Cosine gap and retrieval against normalized layer depth for the six models, baseline mean pooling, test split. Solid lines with filled markers: T5 encoders. Dashed lines with hollow markers: embedding-trained models. (a) Raw gap: mean cosine of same-directory pairs minus that of different-directory pairs. (b) Standardized gap: the raw gap divided by the standard deviation of all pairwise test cosines at that layer. (c) Baseline Task A test AUROC. In the middle layers of mT5-base the raw gap rises to 0.21–0.25 while AUROC falls to 0.65, and the standardized gap falls with AUROC. LaTa’s raw gap is negative at layers 3–11. Across all 100 model-layers the standardized gap tracks AUROC at Spearman 0.97, and the raw gap at 0.22.

Model	Operational layer	Most anisotropic layer	PC1 before	PC1 after	Eff. rank before→after
LaTa	7	8	0.956	0.036	1.34→168.40
PhilTa	1	6	0.862	0.039	1.83→155.02
mT5-base	1	5	1.000	0.044	1.00→151.74

Table 8: Most anisotropic layers, by top-PC share on the test split, and operational attribution layers for the three T5 encoders. PC1: top-PC share. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer, on the test split. The main text (Table 2) reads top-PC share on the training split, where LaTa peaks at layer 4 instead of 8; LaTa’s training curve is flat, with a top-PC share of 0.93 to 0.95 over layers 3–11. Operational layers come from the train-only retrieval rule of Appendix I.

Model	Task A layer				Task B layer			
	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	12	1	12	7	1	1	8	12
PhilTa	1	1	9	8	1	1	1	1
mT5-base	12	1	2	1	12	1	1	2
LaBSE	12	12	11	10	11	12	11	11
Qwen3-0.6B	26	22	2	2	28	27	5	5
KaLM-mini	23	23	1	2	23	22	3	1
LaTa (fine-tuned)	12	–	12	–	12	–	12	–
Qwen3-0.6B (fine-tuned)	28	–	27	–	28	–	27	–
KaLM-mini (fine-tuned)	24	–	24	–	24	–	24	–

Table 9: Train-selected layer behind every cell of Tables 6 and 7. For each model and post-processing setting, the Task A layer has the highest training-set AUROC and the Task B layer the highest training-set directory accuracy at rank 1; the test scores in those tables are read at these layers, so no layer is chosen on test. Layer 1 is the first transformer block. The fine-tuned rows below the rule have no SIF setting. Table 21 reads its pre-trained rows at the Base and ABTT layers above the rule and its fine-tuned rows at the layers below it.

Task A: Pairwise Duplicate Detection (main)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaTa	1	0.934	0.970	0.286	0.566
	2	0.563	0.969	0.019	0.554
	3	0.513	0.973	-0.048	0.555
	4	0.500	0.971	-0.067	0.565
	5	0.496	0.972	-0.069	0.572
	6	0.496	0.964	-0.069	0.569
	7	0.498	0.962	-0.072	0.564
	8	0.502	0.965	-0.074	0.558
	9	0.503	0.965	-0.073	0.557
	10	0.505	0.971	-0.072	0.551
	11	0.508	0.973	-0.067	0.550
	12	0.938	0.971	0.237	0.525
PhilTa	1	0.939	0.977	0.231	0.574
	2	0.911	0.973	0.161	0.553
	3	0.601	0.973	0.085	0.549
	4	0.551	0.978	0.063	0.551
	5	0.542	0.979	0.053	0.551
	6	0.541	0.981	0.047	0.551
	7	0.542	0.979	0.046	0.554
	8	0.542	0.984	0.046	0.555
	9	0.539	0.983	0.041	0.540
	10	0.538	0.982	0.040	0.531
	11	0.539	0.982	0.041	0.511
	12	0.911	0.982	0.181	0.487
mT5-base	1	0.822	0.979	0.027	0.550
	2	0.822	0.975	0.026	0.536
	3	0.804	0.968	0.025	0.517
	4	0.799	0.974	0.040	0.502
	5	0.654	0.978	0.219	0.482
	6	0.657	0.978	0.238	0.483
	7	0.656	0.980	0.244	0.482
	8	0.656	0.982	0.249	0.471
	9	0.660	0.982	0.235	0.475
	10	0.659	0.977	0.226	0.459
	11	0.659	0.974	0.213	0.449
	12	0.838	0.972	0.081	0.442

Table 10: Per-layer Task A pairwise duplicate-detection metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 6 (listed in Table 9), and not always the layer with the highest test score in this table.

Task B: Autonomous Routing (main, file-level, τ -thresholded)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaTa	1	0.630	0.910	0.916	0.907	0.738	0.909
	2	0.153	0.905	0.913	0.892	0.439	0.900
	3	0.256	0.882	0.842	0.950	0.477	0.908
	4	0.389	0.893	0.718	0.920	0.513	0.903
	5	0.374	0.892	0.715	0.913	0.502	0.900
	6	0.378	0.884	0.712	0.913	0.503	0.895
	7	0.372	0.875	0.715	0.916	0.501	0.890
	8	0.379	0.884	0.706	0.885	0.502	0.885
	9	0.366	0.886	0.724	0.873	0.501	0.881
	10	0.344	0.873	0.752	0.882	0.498	0.876
	11	0.310	0.867	0.786	0.901	0.490	0.880
	12	0.576	0.890	0.932	0.833	0.710	0.868
PhilTa	1	0.591	0.918	0.904	0.901	0.709	0.911
	2	0.479	0.895	0.867	0.885	0.625	0.892
	3	0.193	0.897	0.889	0.879	0.455	0.890
	4	0.148	0.886	0.923	0.873	0.439	0.881
	5	0.252	0.879	0.796	0.873	0.457	0.876
	6	0.338	0.905	0.700	0.817	0.474	0.872
	7	0.301	0.901	0.746	0.830	0.469	0.874
	8	0.262	0.875	0.808	0.885	0.467	0.879
	9	0.303	0.890	0.762	0.870	0.476	0.882
	10	0.299	0.908	0.780	0.793	0.480	0.865
	11	0.256	0.869	0.824	0.836	0.470	0.857
	12	0.338	0.880	0.898	0.876	0.549	0.879
mT5-base	1	0.135	0.877	0.981	0.947	0.453	0.903
	2	0.110	0.890	0.975	0.910	0.436	0.897
	3	0.161	0.794	0.926	0.941	0.449	0.850
	4	0.269	0.787	0.858	0.882	0.491	0.823
	5	0.422	0.746	0.659	0.864	0.512	0.790
	6	0.636	0.766	0.489	0.876	0.580	0.808
	7	0.594	0.793	0.517	0.851	0.565	0.815
	8	0.557	0.809	0.560	0.842	0.558	0.822
	9	0.364	0.841	0.746	0.814	0.508	0.831
	10	0.314	0.794	0.783	0.876	0.491	0.825
	11	0.297	0.763	0.780	0.854	0.479	0.797
	12	0.198	0.834	0.926	0.799	0.472	0.821

Table 11: Per-layer Task B autonomous routing accuracy for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 9), and not always the layer with the highest test score in this table.

Task B: Top-K Ranking (main, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaTa	1	0.721	0.880	0.630	0.910	0.916	0.907
	2	0.383	0.869	0.153	0.905	0.913	0.892
	3	0.350	0.882	0.256	0.882	0.842	0.950
	4	0.303	0.881	0.389	0.893	0.718	0.920
	5	0.297	0.875	0.374	0.892	0.715	0.913
	6	0.297	0.874	0.378	0.884	0.712	0.913
	7	0.302	0.868	0.372	0.875	0.715	0.916
	8	0.302	0.861	0.379	0.884	0.706	0.885
	9	0.309	0.858	0.366	0.886	0.724	0.873
	10	0.317	0.852	0.344	0.873	0.752	0.882
	11	0.331	0.857	0.310	0.867	0.786	0.901
	12	0.689	0.841	0.576	0.890	0.932	0.833
PhilTa	1	0.693	0.883	0.591	0.918	0.904	0.901
	2	0.603	0.869	0.479	0.895	0.867	0.885
	3	0.389	0.865	0.193	0.897	0.889	0.879
	4	0.369	0.859	0.148	0.886	0.923	0.873
	5	0.334	0.854	0.252	0.879	0.796	0.873
	6	0.310	0.845	0.338	0.905	0.700	0.817
	7	0.325	0.847	0.301	0.901	0.746	0.830
	8	0.350	0.857	0.262	0.875	0.808	0.885
	9	0.337	0.857	0.303	0.890	0.762	0.870
	10	0.344	0.833	0.299	0.908	0.780	0.793
	11	0.355	0.829	0.256	0.869	0.824	0.836
	12	0.528	0.854	0.338	0.880	0.898	0.876
mT5-base	1	0.451	0.885	0.135	0.877	0.981	0.947
	2	0.435	0.874	0.110	0.890	0.975	0.910
	3	0.445	0.830	0.161	0.794	0.926	0.941
	4	0.467	0.798	0.269	0.787	0.858	0.882
	5	0.352	0.756	0.422	0.746	0.659	0.864
	6	0.346	0.775	0.636	0.766	0.489	0.876
	7	0.352	0.775	0.594	0.793	0.517	0.851
	8	0.366	0.784	0.557	0.809	0.560	0.842
	9	0.394	0.787	0.364	0.841	0.746	0.814
	10	0.387	0.788	0.314	0.794	0.783	0.876
	11	0.378	0.765	0.297	0.763	0.780	0.854
	12	0.466	0.779	0.198	0.834	0.926	0.799

Table 12: Per-layer Task B top- k ranking metrics for the three T5 encoders. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 20 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 9), and not always the layer with the highest test score in this table.

Table 13: Mean versus last-token pooling under ABTT with D tuned per layer on the train split, for the five paper models whose last-token embeddings were extracted (mT5-base was not part of this run). For each model and pooling the layer is the one with the highest training-set directory accuracy at rank 1, the rule behind the ABTT cells of Table 7 (their layers are listed in Table 9); D is the number of principal components removed at that layer. The three score columns are test-set Task B assignment accuracy, Task B directory accuracy at rank 1, and Task A cosine gap at that layer.

Model	Pooling	Layer	D	Assignment acc.	Dir. acc. @1	Cosine gap
LaTa	Mean	8	10	0.885	0.861	0.558
	Last token	7	10	0.664	0.466	0.308
PhilTa	Mean	1	10	0.911	0.883	0.573
	Last token	11	10	0.589	0.406	0.205
LaBSE	Mean	11	10	0.908	0.885	0.610
	Last token	12	10	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	10	0.915	0.894	0.545
	Last token	26	5	0.881	0.864	0.580
KaLM-mini	Mean	3	10	0.917	0.894	0.552
	Last token	24	10	0.875	0.847	0.483

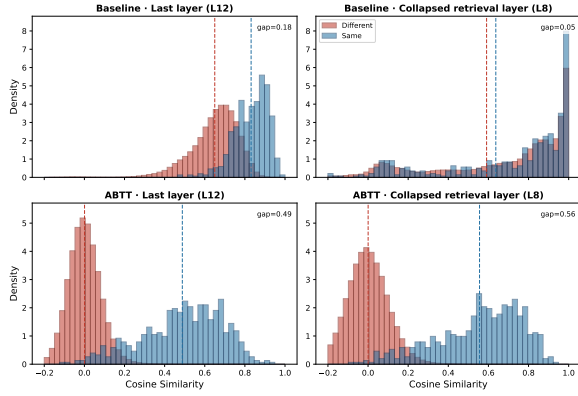


Figure 6: Cosine similarity distributions for PhilTa at the last layer and at its collapsed retrieval layer, before and after ABTT. Equivalent pairs share a directory (Same in the legend); non-equivalent pairs do not (Different). The collapsed retrieval layer is layer 8, chosen by a train-only rule (highest training-set AUROC within 30 to 70 percent of encoder depth, layers 4 to 8), so the panel understates the collapse rather than selecting its worst case; its test AUROC is 0.542, close to chance. Two other PhilTa layers carry other meanings and are not shown here: the most anisotropic layer under the top-PC rule of Appendix B is layer 6, and the lowest baseline test AUROC, 0.538, falls at layer 10. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

Table 14: Per-layer Task A pairwise metrics for the non-T5 models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as the main paper. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $n \times n$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 6 (listed in Table 9), and not always the layer with the highest test score in this table.

Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaBSE	1	0.806	0.977	0.060	0.520
	2	0.833	0.979	0.037	0.531
	3	0.853	0.978	0.062	0.543
	4	0.862	0.977	0.073	0.525
	5	0.864	0.968	0.066	0.501
	6	0.894	0.976	0.078	0.532
	7	0.887	0.975	0.088	0.524
	8	0.879	0.975	0.080	0.522
	9	0.916	0.979	0.090	0.565
	10	0.934	0.982	0.103	0.586
	11	0.960	0.987	0.061	0.611
	12	0.956	0.981	0.183	0.592
Qwen3-0.6B	1	0.858	0.978	0.033	0.564
	2	0.863	0.973	0.024	0.553
	3	0.862	0.980	0.037	0.556
	4	0.860	0.986	0.036	0.547
	5	0.861	0.981	0.030	0.545
	6	0.867	0.983	0.035	0.545
	7	0.876	0.980	0.033	0.544
	8	0.882	0.972	0.037	0.532
	9	0.881	0.977	0.034	0.532
	10	0.887	0.976	0.031	0.529
	11	0.895	0.977	0.030	0.530
	12	0.901	0.973	0.027	0.530
	13	0.907	0.974	0.026	0.523
	14	0.905	0.974	0.024	0.519
	15	0.901	0.971	0.022	0.514
	16	0.910	0.975	0.026	0.509
	17	0.916	0.979	0.032	0.521
	18	0.933	0.983	0.032	0.537
	19	0.936	0.980	0.035	0.541
	20	0.943	0.982	0.041	0.559
	21	0.953	0.982	0.047	0.562
	22	0.963	0.983	0.044	0.583
	23	0.961	0.984	0.038	0.578
	24	0.963	0.984	0.032	0.576
	25	0.961	0.983	0.027	0.578
	26	0.966	0.985	0.024	0.568
	27	0.967	0.982	0.017	0.559
	28	0.958	0.975	0.098	0.556
KaLM-mini	1	0.887	0.981	0.063	0.568
	2	0.906	0.983	0.044	0.566
	3	0.881	0.981	0.062	0.551
	4	0.871	0.983	0.060	0.553
	5	0.861	0.983	0.062	0.541
	6	0.872	0.983	0.063	0.529
	7	0.888	0.979	0.072	0.524
	8	0.890	0.976	0.072	0.524
	9	0.892	0.974	0.066	0.507
	10	0.893	0.971	0.067	0.494
	11	0.894	0.970	0.061	0.475
	12	0.905	0.970	0.071	0.475
	13	0.900	0.968	0.066	0.467

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Task A: Pairwise Duplicate Detection (appendix models)					
Model	Layer	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
	14	0.904	0.971	0.063	0.471
	15	0.907	0.973	0.067	0.468
	16	0.917	0.972	0.071	0.489
	17	0.926	0.975	0.075	0.501
	18	0.945	0.977	0.060	0.519
	19	0.956	0.979	0.061	0.531
	20	0.960	0.980	0.054	0.555
	21	0.962	0.981	0.043	0.555
	22	0.971	0.985	0.050	0.555
	23	0.972	0.986	0.056	0.552
	24	0.956	0.981	0.243	0.520

Table 15: Per-layer Task B autonomous routing accuracy for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1 -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 9), and not always the layer with the highest test score in this table.

Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaBSE	1	0.222	0.879	0.898	0.873	0.477	0.876
	2	0.381	0.875	0.811	0.827	0.543	0.857
	3	0.277	0.886	0.907	0.867	0.514	0.879
	4	0.372	0.897	0.885	0.836	0.565	0.874
	5	0.327	0.845	0.901	0.851	0.543	0.847
	6	0.548	0.860	0.882	0.851	0.674	0.857
	7	0.398	0.849	0.892	0.830	0.584	0.841
	8	0.385	0.869	0.870	0.743	0.568	0.822
	9	0.421	0.871	0.929	0.817	0.612	0.851
	10	0.606	0.892	0.892	0.805	0.713	0.859
	11	0.785	0.923	0.913	0.882	0.833	0.908
	12	0.729	0.914	0.938	0.864	0.808	0.895
Qwen3-0.6B	1	0.264	0.921	0.960	0.885	0.526	0.908
	2	0.437	0.869	0.814	0.916	0.579	0.887
	3	0.250	0.914	0.947	0.907	0.513	0.911
	4	0.411	0.936	0.848	0.842	0.576	0.901
	5	0.181	0.910	0.932	0.923	0.464	0.915
	6	0.422	0.935	0.830	0.796	0.576	0.882
	7	0.250	0.890	0.947	0.876	0.513	0.885
	8	0.409	0.931	0.895	0.799	0.592	0.881
	9	0.460	0.901	0.851	0.851	0.607	0.882
	10	0.533	0.892	0.836	0.864	0.647	0.881
	11	0.293	0.903	0.978	0.808	0.551	0.867
	12	0.407	0.912	0.938	0.817	0.607	0.876
	13	0.432	0.908	0.944	0.836	0.625	0.881
	14	0.456	0.907	0.926	0.811	0.633	0.871
	15	0.469	0.901	0.920	0.845	0.639	0.880
	16	0.639	0.897	0.830	0.808	0.711	0.864
	17	0.480	0.886	0.935	0.851	0.652	0.873
	18	0.553	0.882	0.944	0.839	0.700	0.866
	19	0.514	0.871	0.966	0.858	0.684	0.866
	20	0.654	0.865	0.898	0.892	0.746	0.875
	21	0.647	0.912	0.950	0.898	0.761	0.907
	22	0.731	0.888	0.938	0.916	0.809	0.899
	23	0.622	0.914	0.966	0.895	0.752	0.907

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Task B: Autonomous Routing (appendix models)							
Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
	24	0.748	0.901	0.935	0.935	0.818	0.914
	25	0.574	0.893	0.975	0.923	0.725	0.904
	26	0.613	0.899	0.975	0.895	0.749	0.897
	27	0.793	0.877	0.833	0.923	0.808	0.894
	28	0.796	0.877	0.867	0.913	0.823	0.890
KaLM-mini	1	0.391	0.921	0.907	0.920	0.585	0.921
	2	0.493	0.916	0.848	0.879	0.627	0.902
	3	0.443	0.923	0.870	0.907	0.604	0.917
	4	0.308	0.931	0.935	0.895	0.544	0.917
	5	0.338	0.905	0.882	0.923	0.543	0.911
	6	0.256	0.895	0.910	0.920	0.502	0.904
	7	0.475	0.864	0.854	0.944	0.618	0.894
	8	0.467	0.862	0.870	0.913	0.619	0.881
	9	0.469	0.873	0.885	0.870	0.626	0.872
	10	0.449	0.875	0.885	0.870	0.613	0.873
	11	0.458	0.862	0.885	0.864	0.619	0.862
	12	0.507	0.862	0.895	0.904	0.653	0.878
	13	0.540	0.860	0.895	0.870	0.674	0.864
	14	0.593	0.836	0.845	0.885	0.688	0.854
	15	0.566	0.864	0.898	0.827	0.691	0.850
	16	0.550	0.849	0.935	0.830	0.695	0.841
	17	0.675	0.864	0.920	0.889	0.767	0.873
	18	0.697	0.875	0.960	0.898	0.796	0.883
	19	0.721	0.873	0.966	0.892	0.814	0.880
	20	0.705	0.791	0.966	0.913	0.803	0.837
	21	0.727	0.811	0.966	0.926	0.817	0.854
	22	0.850	0.879	0.916	0.923	0.875	0.895
	23	0.849	0.884	0.920	0.898	0.875	0.889
	24	0.787	0.892	0.944	0.867	0.846	0.882

Table 16: Per-layer Task B top- k ranking metrics for the non-T5 models. baseline is mean pooling without correction; abtt_optimal applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed split (no M -seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 20 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under ABTT. It is the layer behind the ABTT cell of Table 7 (listed in Table 9), and not always the layer with the highest test score in this table.

Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaBSE	1	0.465	0.851	0.222	0.879	0.898	0.873
	2	0.494	0.830	0.381	0.875	0.811	0.827
	3	0.500	0.848	0.277	0.886	0.907	0.867
	4	0.542	0.841	0.372	0.897	0.885	0.836
	5	0.521	0.811	0.327	0.845	0.901	0.851
	6	0.640	0.831	0.548	0.860	0.882	0.851
	7	0.561	0.810	0.398	0.849	0.892	0.830
	8	0.544	0.779	0.385	0.869	0.870	0.743
	9	0.608	0.823	0.421	0.871	0.929	0.817
	10	0.693	0.832	0.606	0.892	0.892	0.805
	11	0.815	0.885	0.785	0.923	0.913	0.882
	12	0.791	0.868	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.521	0.875	0.264	0.921	0.960	0.885
	2	0.558	0.871	0.437	0.869	0.814	0.916
	3	0.508	0.890	0.250	0.914	0.947	0.907
	4	0.559	0.874	0.411	0.936	0.848	0.842
	5	0.450	0.894	0.181	0.910	0.932	0.923
	6	0.551	0.858	0.422	0.935	0.830	0.796
	7	0.507	0.859	0.250	0.890	0.947	0.876

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Task B: Top-K Ranking (appendix models, single-seed)							
Model	Layer	Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
	8	0.580	0.852	0.409	0.931	0.895	0.799
	9	0.587	0.852	0.460	0.901	0.851	0.851
	10	0.622	0.848	0.533	0.892	0.836	0.864
	11	0.548	0.830	0.293	0.903	0.978	0.808
	12	0.603	0.848	0.407	0.912	0.938	0.817
	13	0.618	0.851	0.432	0.908	0.944	0.836
	14	0.621	0.840	0.456	0.907	0.926	0.811
	15	0.627	0.848	0.469	0.901	0.920	0.845
	16	0.682	0.830	0.639	0.897	0.830	0.808
	17	0.641	0.840	0.480	0.886	0.935	0.851
	18	0.691	0.837	0.553	0.882	0.944	0.839
	19	0.676	0.837	0.514	0.871	0.966	0.858
	20	0.733	0.852	0.654	0.865	0.898	0.892
	21	0.752	0.881	0.647	0.912	0.950	0.898
	22	0.795	0.876	0.731	0.888	0.938	0.916
	23	0.742	0.886	0.622	0.914	0.966	0.895
	24	0.808	0.896	0.748	0.901	0.935	0.935
	25	0.718	0.886	0.574	0.893	0.975	0.923
	26	0.740	0.881	0.613	0.899	0.975	0.895
	27	0.787	0.876	0.793	0.877	0.833	0.923
	28	0.803	0.872	0.796	0.877	0.867	0.913
KaLM-mini	1	0.577	0.902	0.391	0.921	0.907	0.920
	2	0.607	0.879	0.493	0.916	0.848	0.879
	3	0.587	0.894	0.443	0.923	0.870	0.907
	4	0.538	0.887	0.308	0.931	0.935	0.895
	5	0.524	0.882	0.338	0.905	0.882	0.923
	6	0.490	0.876	0.256	0.895	0.910	0.920
	7	0.592	0.873	0.475	0.864	0.854	0.944
	8	0.597	0.860	0.467	0.862	0.870	0.913
	9	0.601	0.843	0.469	0.873	0.885	0.870
	10	0.584	0.844	0.449	0.875	0.885	0.870
	11	0.586	0.836	0.458	0.862	0.885	0.864
	12	0.622	0.850	0.507	0.862	0.895	0.904
	13	0.633	0.840	0.540	0.860	0.895	0.870
	14	0.636	0.829	0.593	0.836	0.845	0.885
	15	0.650	0.817	0.566	0.864	0.898	0.827
	16	0.676	0.819	0.550	0.849	0.935	0.830
	17	0.747	0.852	0.675	0.864	0.920	0.889
	18	0.777	0.865	0.697	0.875	0.960	0.898
	19	0.801	0.861	0.721	0.873	0.966	0.892
	20	0.789	0.817	0.705	0.791	0.966	0.913
	21	0.802	0.831	0.727	0.811	0.966	0.926
	22	0.857	0.875	0.850	0.879	0.916	0.923
	23	0.859	0.872	0.849	0.884	0.920	0.898
	24	0.833	0.858	0.787	0.892	0.944	0.867

G SIF and Whitening Post-Processing Sweeps

The main text uses ABTT-only as the paper-facing repair. This appendix reports SIF-only, SIF plus fixed- D ABTT, and SIF plus tuned- D ABTT, and gives the full account of the PCA whitening variant that Section 3 excludes. Table 17 first gives the five-seed cumulative top- K result discussed in the main text, with SIF plus ABTT at each model’s train-selected layer. The per-layer tables then keep one primary metric per task so the four-method grid stays readable.

Why whitening is excluded. PCA whitening (Su et al., 2021) leaves the different-directory cosine mean at or below 0.015, so the learned threshold degenerates: across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, new accuracy never exceeds one file in 323, and assignment accuracy sits at the 62.4 percent class prior. That prior beats baseline in 60 of 100 cells while DirAcc@1 never exceeds 41.5 percent, so the apparent gain is a thresholding artifact. SIF and whitening fall short of ABTT for different reasons. SIF acts before pooling, so it can only attenuate the frequent tokens that feed the dominant direction, and whatever survives that reweighting is still pooled into the passage vector; whitening acts after pooling but rescales all directions instead of deleting the top few, which flattens the same-directory margin along with the common component and so degenerates the routing threshold.

Top- k ranking. In the top- k ranking form of Task B, each directory receives its best document match score, a pseudo-option NEW receives score τ , and the simulated expert succeeds when the labeled directory appears in the shortlist. Cumulative top- K accuracy is the fraction of test queries whose labeled directory is among the top K options. Table 17 gives the shortlist view (per-layer: Appendix D): every model exceeds 95% by Top-2 and gains little after Top-3.

Model	Top-1	Top-2	Top-3	Top-4	Top-5
LaTa	88.9 $\pm$ 0.5	96.1 $\pm$ 0.5	97.4 $\pm$ 0.6	97.9 $\pm$ 0.3	98.2 $\pm$ 0.2
PhilTa	89.8 $\pm$ 0.8	95.8 $\pm$ 0.6	97.4 $\pm$ 0.3	97.7 $\pm$ 0.3	97.8 $\pm$ 0.4
mT5-base	89.3 $\pm$ 0.5	95.1 $\pm$ 0.2	96.7 $\pm$ 0.7	97.1 $\pm$ 0.4	97.2 $\pm$ 0.4
LaBSE	90.6 $\pm$ 1.0	95.5 $\pm$ 0.5	96.4 $\pm$ 0.5	97.2 $\pm$ 0.6	97.8 $\pm$ 0.5
Qwen3-0.6B	89.3 $\pm$ 0.7	95.7 $\pm$ 0.4	97.1 $\pm$ 0.3	97.5 $\pm$ 0.4	97.6 $\pm$ 0.4
KaLM-mini	89.3 $\pm$ 0.5	95.1 $\pm$ 0.4	96.6 $\pm$ 0.7	97.2 $\pm$ 0.7	97.6 $\pm$ 0.5

Table 17: Cumulative top- K accuracy for Task B under SIF+ABTT, as mean $\pm$ standard deviation over five query/reference reseeds at a fixed split. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 7, listed in Table 9. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean. Each row is the bold row of Table 20. Each cell is the percentage of test queries whose labelled directory appears within the top K options. Per-layer top- k rankings: Appendix Tables 12, 16, and 20.

Table 18: Per-layer Task A AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed $D=10$; `sif_abtt_optimal` tunes D per layer on the train split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 10 and 14. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set AUROC under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 6 (listed in Table 9), and not always the layer with the highest test score in this table.

Task A: SIF-suite AUROC (appendix)					
Model	Layer	AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
LaTa	1	0.934	0.948	0.972	0.976
	2	0.563	0.938	0.964	0.971
	3	0.513	0.921	0.967	0.966
	4	0.500	0.890	0.970	0.968
	5	0.496	0.878	0.970	0.970
	6	0.496	0.865	0.965	0.965
	7	0.498	0.858	0.966	0.966
	8	0.502	0.842	0.964	0.964
	9	0.503	0.854	0.964	0.964
	10	0.505	0.859	0.966	0.966
	11	0.508	0.867	0.965	0.965
	12	0.938	0.948	0.963	0.963
PhilTa	1	0.939	0.955	0.975	0.975
	2	0.911	0.941	0.973	0.973
	3	0.601	0.928	0.968	0.968
	4	0.551	0.916	0.972	0.972
	5	0.542	0.904	0.972	0.972
	6	0.541	0.884	0.981	0.981
	7	0.542	0.891	0.981	0.981
	8	0.542	0.895	0.982	0.982
	9	0.539	0.881	0.977	0.977
	10	0.538	0.874	0.977	0.977
	11	0.539	0.872	0.979	0.979
	12	0.911	0.929	0.969	0.969
mT5-base	1	0.822	0.845	0.974	0.974
	2	0.822	0.835	0.966	0.966
	3	0.804	0.808	0.958	0.958
	4	0.799	0.809	0.966	0.966
	5	0.654	0.672	0.974	0.974
	6	0.657	0.678	0.971	0.971
	7	0.656	0.680	0.967	0.967
	8	0.656	0.681	0.968	0.968
	9	0.660	0.685	0.967	0.967
	10	0.659	0.684	0.966	0.966
	11	0.659	0.689	0.966	0.966
	12	0.838	0.833	0.970	0.970
LaBSE	1	0.806	0.818	0.974	0.974
	2	0.833	0.840	0.975	0.975
	3	0.853	0.862	0.968	0.968
	4	0.862	0.869	0.966	0.966
	5	0.864	0.869	0.964	0.964
	6	0.894	0.895	0.965	0.965
	7	0.887	0.892	0.964	0.964
	8	0.879	0.883	0.968	0.968
	9	0.916	0.915	0.973	0.973
	10	0.934	0.934	0.976	0.976
	11	0.960	0.959	0.983	0.983
	12	0.956	0.956	0.980	0.984
Qwen3-0.6B	1	0.858	0.852	0.982	0.981
	2	0.863	0.853	0.972	0.972
	3	0.862	0.858	0.977	0.975
	4	0.860	0.860	0.976	0.977
	5	0.861	0.858	0.971	0.971
	6	0.867	0.861	0.974	0.974
	7	0.876	0.866	0.971	0.971

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Task A: SIF-suite AUROC (appendix)					
Model	Layer	AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT ₁₀)	AUROC (SIF+ABTT)
	8	0.882	0.868	0.969	0.969
	9	0.881	0.866	0.968	0.968
	10	0.887	0.870	0.966	0.966
	11	0.895	0.875	0.964	0.964
	12	0.901	0.881	0.966	0.966
	13	0.907	0.887	0.967	0.967
	14	0.905	0.885	0.966	0.966
	15	0.901	0.883	0.968	0.968
	16	0.910	0.894	0.965	0.965
	17	0.916	0.899	0.967	0.974
	18	0.933	0.914	0.966	0.973
	19	0.936	0.915	0.966	0.976
	20	0.943	0.918	0.971	0.977
	21	0.953	0.931	0.971	0.971
	22	0.963	0.942	0.973	0.973
	23	0.961	0.941	0.973	0.973
	24	0.963	0.943	0.976	0.974
	25	0.961	0.938	0.978	0.975
	26	0.966	0.942	0.980	0.980
	27	0.967	0.936	0.978	0.978
	28	0.958	0.925	0.968	0.968
KaLM-mini	1	0.887	0.880	0.982	0.980
	2	0.906	0.899	0.981	0.981
	3	0.881	0.882	0.977	0.977
	4	0.871	0.873	0.974	0.974
	5	0.861	0.862	0.971	0.971
	6	0.872	0.867	0.967	0.967
	7	0.888	0.879	0.972	0.972
	8	0.890	0.880	0.973	0.975
	9	0.892	0.882	0.968	0.968
	10	0.893	0.882	0.964	0.964
	11	0.894	0.885	0.969	0.969
	12	0.905	0.896	0.966	0.966
	13	0.900	0.894	0.964	0.964
	14	0.904	0.898	0.970	0.970
	15	0.907	0.901	0.970	0.970
	16	0.917	0.913	0.967	0.967
	17	0.926	0.923	0.968	0.968
	18	0.945	0.939	0.977	0.977
	19	0.956	0.950	0.978	0.978
	20	0.960	0.956	0.978	0.978
	21	0.962	0.958	0.980	0.983
	22	0.971	0.967	0.983	0.984
	23	0.972	0.964	0.986	0.985
	24	0.956	0.956	0.976	0.976

Table 19: Per-layer Task B overall assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 11 and 15 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the layer chosen on the train split, the layer with the highest training-set directory accuracy at rank 1 under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 7 (listed in Table 9), and not always the layer with the highest test score in this table.

Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
LaTa	1	0.738	0.832	0.920	0.899
	2	0.439	0.791	0.904	0.900
	3	0.477	0.720	0.910	0.907
	4	0.513	0.668	0.900	0.902
	5	0.502	0.648	0.906	0.906
	6	0.503	0.668	0.895	0.895

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Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
	7	0.501	0.642	0.903	0.903
	8	0.502	0.614	0.892	0.892
	9	0.501	0.618	0.889	0.889
	10	0.498	0.655	0.890	0.890
	11	0.490	0.621	0.880	0.880
	12	0.710	0.804	0.902	0.902
PhilTa	1	0.709	0.790	0.914	0.914
	2	0.625	0.738	0.892	0.892
	3	0.455	0.713	0.892	0.892
	4	0.439	0.663	0.886	0.886
	5	0.457	0.626	0.899	0.899
	6	0.474	0.612	0.906	0.906
	7	0.469	0.625	0.906	0.906
	8	0.467	0.614	0.886	0.886
	9	0.476	0.590	0.895	0.895
	10	0.480	0.580	0.888	0.888
	11	0.470	0.565	0.886	0.886
	12	0.549	0.667	0.899	0.899
mT5-base	1	0.453	0.545	0.907	0.907
	2	0.436	0.537	0.903	0.903
	3	0.449	0.568	0.885	0.885
	4	0.491	0.488	0.857	0.857
	5	0.512	0.483	0.826	0.826
	6	0.580	0.558	0.817	0.817
	7	0.565	0.554	0.832	0.832
	8	0.558	0.548	0.834	0.834
	9	0.508	0.495	0.845	0.845
	10	0.491	0.495	0.832	0.832
	11	0.479	0.484	0.816	0.816
	12	0.472	0.541	0.858	0.858
LaBSE	1	0.477	0.521	0.886	0.886
	2	0.543	0.508	0.875	0.875
	3	0.514	0.577	0.860	0.860
	4	0.565	0.587	0.869	0.869
	5	0.543	0.598	0.857	0.857
	6	0.674	0.648	0.872	0.872
	7	0.584	0.634	0.858	0.858
	8	0.568	0.571	0.854	0.854
	9	0.612	0.613	0.859	0.859
	10	0.713	0.698	0.864	0.864
	11	0.833	0.807	0.896	0.896
	12	0.808	0.834	0.890	0.880
Qwen3-0.6B	1	0.526	0.566	0.914	0.909
	2	0.579	0.524	0.894	0.894
	3	0.513	0.537	0.904	0.901
	4	0.576	0.531	0.899	0.904
	5	0.464	0.537	0.908	0.908
	6	0.576	0.526	0.906	0.906
	7	0.513	0.573	0.911	0.911
	8	0.592	0.627	0.911	0.911
	9	0.607	0.576	0.910	0.910
	10	0.647	0.599	0.900	0.900
	11	0.551	0.606	0.893	0.893
	12	0.607	0.663	0.900	0.900
	13	0.625	0.675	0.897	0.897
	14	0.633	0.571	0.893	0.893
	15	0.639	0.600	0.893	0.893
	16	0.711	0.679	0.880	0.880
	17	0.652	0.686	0.900	0.876
	18	0.700	0.731	0.899	0.882
	19	0.684	0.700	0.886	0.850
	20	0.746	0.676	0.882	0.866

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Task B: SIF-suite routing (appendix)					
Model	Layer	Overall (base)	Overall (SIF)	Overall (SIF+ABTT ₁₀)	Overall (SIF+ABTT)
KaLM-mini	21	0.761	0.739	0.892	0.892
	22	0.809	0.719	0.893	0.893
	23	0.752	0.759	0.893	0.893
	24	0.818	0.706	0.900	0.899
	25	0.725	0.748	0.897	0.893
	26	0.749	0.772	0.903	0.903
	27	0.808	0.760	0.899	0.899
	28	0.823	0.774	0.890	0.890
	1	0.585	0.604	0.906	0.908
	2	0.627	0.594	0.908	0.908
	3	0.604	0.576	0.907	0.907
	4	0.544	0.577	0.901	0.901
	5	0.543	0.533	0.897	0.897
	6	0.502	0.558	0.900	0.900
	7	0.618	0.593	0.901	0.901
	8	0.619	0.607	0.892	0.878
	9	0.626	0.663	0.876	0.876
	10	0.613	0.603	0.866	0.866
	11	0.619	0.627	0.874	0.874
	12	0.653	0.642	0.874	0.874
	13	0.674	0.666	0.864	0.864
	14	0.688	0.648	0.861	0.861
	15	0.691	0.698	0.866	0.866
	16	0.695	0.738	0.859	0.859
	17	0.767	0.763	0.882	0.882
	18	0.796	0.796	0.879	0.879
	19	0.814	0.804	0.878	0.878
	20	0.803	0.821	0.876	0.876
	21	0.817	0.832	0.874	0.876
	22	0.875	0.857	0.876	0.876
	23	0.875	0.862	0.880	0.881
	24	0.846	0.847	0.869	0.869

Table 20: Per-layer Task B top- k ranking accuracy across all six models, averaged over $M = 5$ query/reference reseeds of the train/test split (mean $\pm$ std). `sif_abtt_optimal` applies SIF weighting plus ABTT with D tuned per layer on the train split; the mseed sweep was not run for the ABTT-only variant of Tables 6 and 7, so we include this table to give a variance-aware view of the SIF-conditioned best variant. Overall assignment accuracy coincides with Acc@1 in the mseed pipeline, so we report the existing/new decomposition instead. Rows in bold mark the layer reported per model, the row printed in Table 17. For each model the reported layer is chosen on the train split: the layer with the highest training-set directory accuracy at rank 1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 7, listed in Table 9. D is tuned per layer on the train split. This is not always the layer with the highest five-seed mean.

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
LaTa	1	0.731 $\pm$ 0.010	0.889 $\pm$ 0.010	0.563 $\pm$ 0.007	0.844 $\pm$ 0.011	0.949 $\pm$ 0.011	0.948 $\pm$ 0.010
	2	0.429 $\pm$ 0.011	0.888 $\pm$ 0.007	0.049 $\pm$ 0.006	0.841 $\pm$ 0.007	0.920 $\pm$ 0.008	0.948 $\pm$ 0.007
	3	0.393 $\pm$ 0.011	0.891 $\pm$ 0.010	0.047 $\pm$ 0.006	0.847 $\pm$ 0.010	0.840 $\pm$ 0.020	0.947 $\pm$ 0.012
	4	0.331 $\pm$ 0.019	0.889 $\pm$ 0.009	0.048 $\pm$ 0.009	0.860 $\pm$ 0.009	0.696 $\pm$ 0.030	0.925 $\pm$ 0.013
	5	0.326 $\pm$ 0.021	0.891 $\pm$ 0.009	0.038 $\pm$ 0.005	0.858 $\pm$ 0.008	0.698 $\pm$ 0.028	0.934 $\pm$ 0.014
	6	0.322 $\pm$ 0.018	0.891 $\pm$ 0.009	0.038 $\pm$ 0.007	0.863 $\pm$ 0.010	0.688 $\pm$ 0.026	0.927 $\pm$ 0.013
	7	0.328 $\pm$ 0.020	0.896 $\pm$ 0.008	0.045 $\pm$ 0.007	0.848 $\pm$ 0.009	0.693 $\pm$ 0.028	0.957 $\pm$ 0.010
	8	0.325 $\pm$ 0.022	0.884 $\pm$ 0.008	0.046 $\pm$ 0.010	0.834 $\pm$ 0.008	0.685 $\pm$ 0.032	0.949 $\pm$ 0.011
	9	0.332 $\pm$ 0.019	0.886 $\pm$ 0.011	0.045 $\pm$ 0.010	0.848 $\pm$ 0.009	0.701 $\pm$ 0.024	0.935 $\pm$ 0.015
	10	0.346 $\pm$ 0.017	0.889 $\pm$ 0.004	0.045 $\pm$ 0.010	0.846 $\pm$ 0.008	0.735 $\pm$ 0.021	0.944 $\pm$ 0.012
	11	0.361 $\pm$ 0.018	0.882 $\pm$ 0.009	0.045 $\pm$ 0.009	0.862 $\pm$ 0.010	0.767 $\pm$ 0.029	0.907 $\pm$ 0.014
	12	0.707 $\pm$ 0.014	0.889 $\pm$ 0.005	0.511 $\pm$ 0.016	0.834 $\pm$ 0.005	0.961 $\pm$ 0.008	0.961 $\pm$ 0.010
PhilTa	1	0.705 $\pm$ 0.015	0.898 $\pm$ 0.008	0.531 $\pm$ 0.016	0.858 $\pm$ 0.010	0.931 $\pm$ 0.012	0.951 $\pm$ 0.011

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Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	2	0.626 $\pm$ 0.019	0.888 $\pm$ 0.010	0.410 $\pm$ 0.017	0.866 $\pm$ 0.010	0.906 $\pm$ 0.018	0.917 $\pm$ 0.017
	3	0.440 $\pm$ 0.013	0.893 $\pm$ 0.009	0.071 $\pm$ 0.014	0.858 $\pm$ 0.010	0.917 $\pm$ 0.013	0.939 $\pm$ 0.011
	4	0.421 $\pm$ 0.016	0.891 $\pm$ 0.009	0.027 $\pm$ 0.007	0.858 $\pm$ 0.008	0.928 $\pm$ 0.015	0.935 $\pm$ 0.013
	5	0.375 $\pm$ 0.016	0.895 $\pm$ 0.010	0.042 $\pm$ 0.013	0.852 $\pm$ 0.010	0.804 $\pm$ 0.018	0.950 $\pm$ 0.014
	6	0.337 $\pm$ 0.015	0.897 $\pm$ 0.009	0.060 $\pm$ 0.012	0.861 $\pm$ 0.010	0.696 $\pm$ 0.026	0.944 $\pm$ 0.013
	7	0.361 $\pm$ 0.017	0.900 $\pm$ 0.009	0.056 $\pm$ 0.012	0.860 $\pm$ 0.008	0.755 $\pm$ 0.020	0.952 $\pm$ 0.013
	8	0.389 $\pm$ 0.016	0.891 $\pm$ 0.010	0.058 $\pm$ 0.014	0.859 $\pm$ 0.009	0.818 $\pm$ 0.018	0.932 $\pm$ 0.015
	9	0.371 $\pm$ 0.016	0.893 $\pm$ 0.008	0.066 $\pm$ 0.014	0.861 $\pm$ 0.009	0.766 $\pm$ 0.027	0.934 $\pm$ 0.016
	10	0.377 $\pm$ 0.014	0.892 $\pm$ 0.009	0.066 $\pm$ 0.014	0.855 $\pm$ 0.005	0.778 $\pm$ 0.023	0.940 $\pm$ 0.017
	11	0.394 $\pm$ 0.014	0.888 $\pm$ 0.008	0.059 $\pm$ 0.010	0.851 $\pm$ 0.010	0.827 $\pm$ 0.021	0.935 $\pm$ 0.013
	12	0.566 $\pm$ 0.013	0.897 $\pm$ 0.010	0.282 $\pm$ 0.014	0.863 $\pm$ 0.010	0.934 $\pm$ 0.007	0.941 $\pm$ 0.018
mT5-base	1	0.487 $\pm$ 0.012	0.899 $\pm$ 0.005	0.100 $\pm$ 0.009	0.856 $\pm$ 0.007	0.987 $\pm$ 0.007	0.954 $\pm$ 0.011
	2	0.476 $\pm$ 0.012	0.893 $\pm$ 0.005	0.083 $\pm$ 0.008	0.851 $\pm$ 0.012	0.984 $\pm$ 0.008	0.947 $\pm$ 0.013
	3	0.483 $\pm$ 0.014	0.883 $\pm$ 0.011	0.124 $\pm$ 0.009	0.817 $\pm$ 0.014	0.947 $\pm$ 0.008	0.970 $\pm$ 0.007
	4	0.500 $\pm$ 0.015	0.852 $\pm$ 0.007	0.206 $\pm$ 0.006	0.772 $\pm$ 0.010	0.879 $\pm$ 0.020	0.956 $\pm$ 0.010
	5	0.371 $\pm$ 0.014	0.825 $\pm$ 0.008	0.155 $\pm$ 0.006	0.724 $\pm$ 0.007	0.651 $\pm$ 0.023	0.956 $\pm$ 0.006
	6	0.361 $\pm$ 0.012	0.810 $\pm$ 0.005	0.263 $\pm$ 0.021	0.765 $\pm$ 0.012	0.488 $\pm$ 0.018	0.868 $\pm$ 0.016
	7	0.370 $\pm$ 0.010	0.827 $\pm$ 0.008	0.255 $\pm$ 0.014	0.760 $\pm$ 0.011	0.519 $\pm$ 0.017	0.914 $\pm$ 0.013
	8	0.382 $\pm$ 0.008	0.824 $\pm$ 0.012	0.244 $\pm$ 0.007	0.773 $\pm$ 0.012	0.561 $\pm$ 0.017	0.891 $\pm$ 0.016
	9	0.428 $\pm$ 0.014	0.838 $\pm$ 0.010	0.171 $\pm$ 0.008	0.783 $\pm$ 0.013	0.759 $\pm$ 0.024	0.909 $\pm$ 0.006
	10	0.421 $\pm$ 0.010	0.821 $\pm$ 0.017	0.140 $\pm$ 0.011	0.767 $\pm$ 0.015	0.783 $\pm$ 0.021	0.892 $\pm$ 0.019
	11	0.420 $\pm$ 0.015	0.803 $\pm$ 0.012	0.131 $\pm$ 0.012	0.721 $\pm$ 0.013	0.792 $\pm$ 0.016	0.910 $\pm$ 0.015
	12	0.507 $\pm$ 0.009	0.841 $\pm$ 0.009	0.169 $\pm$ 0.007	0.772 $\pm$ 0.016	0.942 $\pm$ 0.011	0.931 $\pm$ 0.011
LaBSE	1	0.502 $\pm$ 0.012	0.858 $\pm$ 0.007	0.170 $\pm$ 0.003	0.838 $\pm$ 0.007	0.929 $\pm$ 0.016	0.885 $\pm$ 0.009
	2	0.523 $\pm$ 0.011	0.860 $\pm$ 0.004	0.274 $\pm$ 0.005	0.817 $\pm$ 0.003	0.844 $\pm$ 0.022	0.915 $\pm$ 0.008
	3	0.535 $\pm$ 0.010	0.847 $\pm$ 0.006	0.218 $\pm$ 0.009	0.831 $\pm$ 0.009	0.945 $\pm$ 0.008	0.867 $\pm$ 0.007
	4	0.569 $\pm$ 0.008	0.850 $\pm$ 0.006	0.299 $\pm$ 0.004	0.837 $\pm$ 0.006	0.917 $\pm$ 0.014	0.866 $\pm$ 0.011
	5	0.551 $\pm$ 0.012	0.845 $\pm$ 0.005	0.255 $\pm$ 0.012	0.794 $\pm$ 0.005	0.933 $\pm$ 0.013	0.911 $\pm$ 0.010
	6	0.661 $\pm$ 0.011	0.859 $\pm$ 0.004	0.461 $\pm$ 0.019	0.817 $\pm$ 0.008	0.919 $\pm$ 0.018	0.914 $\pm$ 0.015
	7	0.596 $\pm$ 0.010	0.849 $\pm$ 0.004	0.333 $\pm$ 0.016	0.808 $\pm$ 0.006	0.935 $\pm$ 0.016	0.902 $\pm$ 0.010
	8	0.572 $\pm$ 0.012	0.847 $\pm$ 0.006	0.314 $\pm$ 0.019	0.802 $\pm$ 0.009	0.904 $\pm$ 0.016	0.905 $\pm$ 0.008
	9	0.631 $\pm$ 0.008	0.854 $\pm$ 0.007	0.381 $\pm$ 0.015	0.828 $\pm$ 0.002	0.954 $\pm$ 0.008	0.887 $\pm$ 0.012
	10	0.712 $\pm$ 0.016	0.858 $\pm$ 0.005	0.551 $\pm$ 0.019	0.836 $\pm$ 0.005	0.920 $\pm$ 0.015	0.887 $\pm$ 0.010
	11	0.820 $\pm$ 0.010	0.906 $\pm$ 0.010	0.731 $\pm$ 0.012	0.875 $\pm$ 0.013	0.935 $\pm$ 0.012	0.944 $\pm$ 0.017
	12	0.793 $\pm$ 0.015	0.884 $\pm$ 0.006	0.670 $\pm$ 0.019	0.850 $\pm$ 0.012	0.952 $\pm$ 0.013	0.927 $\pm$ 0.015
Qwen3-0.6B	1	0.550 $\pm$ 0.008	0.894 $\pm$ 0.005	0.223 $\pm$ 0.006	0.840 $\pm$ 0.010	0.972 $\pm$ 0.003	0.965 $\pm$ 0.011
	2	0.572 $\pm$ 0.015	0.882 $\pm$ 0.004	0.363 $\pm$ 0.014	0.828 $\pm$ 0.005	0.841 $\pm$ 0.032	0.952 $\pm$ 0.007
	3	0.541 $\pm$ 0.007	0.889 $\pm$ 0.006	0.216 $\pm$ 0.007	0.827 $\pm$ 0.008	0.962 $\pm$ 0.007	0.968 $\pm$ 0.011
	4	0.580 $\pm$ 0.012	0.891 $\pm$ 0.004	0.353 $\pm$ 0.008	0.845 $\pm$ 0.007	0.874 $\pm$ 0.028	0.950 $\pm$ 0.006
	5	0.487 $\pm$ 0.017	0.893 $\pm$ 0.007	0.133 $\pm$ 0.012	0.839 $\pm$ 0.011	0.944 $\pm$ 0.014	0.964 $\pm$ 0.009
	6	0.568 $\pm$ 0.012	0.897 $\pm$ 0.005	0.350 $\pm$ 0.010	0.857 $\pm$ 0.008	0.850 $\pm$ 0.023	0.948 $\pm$ 0.012
	7	0.541 $\pm$ 0.007	0.902 $\pm$ 0.006	0.211 $\pm$ 0.009	0.862 $\pm$ 0.008	0.967 $\pm$ 0.012	0.953 $\pm$ 0.012
	8	0.609 $\pm$ 0.009	0.903 $\pm$ 0.006	0.362 $\pm$ 0.013	0.861 $\pm$ 0.010	0.929 $\pm$ 0.017	0.956 $\pm$ 0.005
	9	0.612 $\pm$ 0.013	0.900 $\pm$ 0.004	0.393 $\pm$ 0.016	0.868 $\pm$ 0.008	0.894 $\pm$ 0.021	0.943 $\pm$ 0.006
	10	0.638 $\pm$ 0.013	0.890 $\pm$ 0.009	0.452 $\pm$ 0.009	0.847 $\pm$ 0.014	0.878 $\pm$ 0.022	0.944 $\pm$ 0.010
	11	0.578 $\pm$ 0.006	0.885 $\pm$ 0.005	0.257 $\pm$ 0.010	0.838 $\pm$ 0.014	0.991 $\pm$ 0.005	0.947 $\pm$ 0.012
	12	0.627 $\pm$ 0.009	0.892 $\pm$ 0.005	0.366 $\pm$ 0.016	0.844 $\pm$ 0.010	0.964 $\pm$ 0.013	0.955 $\pm$ 0.007
	13	0.639 $\pm$ 0.010	0.887 $\pm$ 0.006	0.383 $\pm$ 0.014	0.849 $\pm$ 0.011	0.970 $\pm$ 0.015	0.937 $\pm$ 0.011
	14	0.646 $\pm$ 0.009	0.885 $\pm$ 0.006	0.402 $\pm$ 0.017	0.852 $\pm$ 0.009	0.960 $\pm$ 0.016	0.927 $\pm$ 0.007
	15	0.648 $\pm$ 0.012	0.885 $\pm$ 0.007	0.410 $\pm$ 0.017	0.852 $\pm$ 0.010	0.954 $\pm$ 0.017	0.929 $\pm$ 0.018
	16	0.691 $\pm$ 0.016	0.873 $\pm$ 0.005	0.546 $\pm$ 0.017	0.837 $\pm$ 0.009	0.878 $\pm$ 0.017	0.920 $\pm$ 0.016
	17	0.658 $\pm$ 0.010	0.862 $\pm$ 0.006	0.420 $\pm$ 0.018	0.807 $\pm$ 0.012	0.966 $\pm$ 0.013	0.933 $\pm$ 0.013
	18	0.709 $\pm$ 0.012	0.868 $\pm$ 0.002	0.503 $\pm$ 0.018	0.833 $\pm$ 0.013	0.975 $\pm$ 0.006	0.914 $\pm$ 0.017
	19	0.691 $\pm$ 0.010	0.874 $\pm$ 0.002	0.462 $\pm$ 0.018	0.860 $\pm$ 0.007	0.986 $\pm$ 0.006	0.893 $\pm$ 0.012
	20	0.743 $\pm$ 0.020	0.860 $\pm$ 0.004	0.588 $\pm$ 0.023	0.797 $\pm$ 0.011	0.943 $\pm$ 0.018	0.941 $\pm$ 0.010
	21	0.758 $\pm$ 0.017	0.879 $\pm$ 0.006	0.585 $\pm$ 0.021	0.836 $\pm$ 0.008	0.981 $\pm$ 0.010	0.935 $\pm$ 0.010
	22	0.799 $\pm$ 0.014	0.882 $\pm$ 0.006	0.671 $\pm$ 0.019	0.855 $\pm$ 0.002	0.965 $\pm$ 0.006	0.918 $\pm$ 0.015
	23	0.757 $\pm$ 0.012	0.887 $\pm$ 0.006	0.577 $\pm$ 0.014	0.857 $\pm$ 0.007	0.989 $\pm$ 0.004	0.927 $\pm$ 0.011
	24	0.810 $\pm$ 0.014	0.893 $\pm$ 0.005	0.695 $\pm$ 0.021	0.861 $\pm$ 0.004	0.959 $\pm$ 0.003	0.935 $\pm$ 0.011
	25	0.732 $\pm$ 0.011	0.890 $\pm$ 0.009	0.527 $\pm$ 0.016	0.866 $\pm$ 0.006	0.997 $\pm$ 0.003	0.920 $\pm$ 0.014
	26	0.749 $\pm$ 0.013	0.895 $\pm$ 0.008	0.562 $\pm$ 0.017	0.858 $\pm$ 0.004	0.991 $\pm$ 0.003	0.943 $\pm$ 0.015
	27	0.812 $\pm$ 0.004	0.881 $\pm$ 0.005	0.741 $\pm$ 0.008	0.824 $\pm$ 0.006	0.903 $\pm$ 0.011	0.954 $\pm$ 0.011

(continued on next page)

Task B: Top-K Ranking (appendix, 5-seed mean $\pm$ std, SIF+ABTT)							
Model	Layer	Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	28	0.824 $\pm$ 0.010	0.871 $\pm$ 0.006	0.736 $\pm$ 0.015	0.802 $\pm$ 0.010	0.939 $\pm$ 0.005	0.961 $\pm$ 0.007
KaLM-mini	1	0.596 $\pm$ 0.012	0.893 $\pm$ 0.005	0.344 $\pm$ 0.003	0.846 $\pm$ 0.005	0.922 $\pm$ 0.022	0.953 $\pm$ 0.006
	2	0.620 $\pm$ 0.018	0.893 $\pm$ 0.003	0.422 $\pm$ 0.014	0.845 $\pm$ 0.005	0.876 $\pm$ 0.023	0.955 $\pm$ 0.008
	3	0.605 $\pm$ 0.011	0.893 $\pm$ 0.004	0.379 $\pm$ 0.008	0.843 $\pm$ 0.000	0.896 $\pm$ 0.018	0.959 $\pm$ 0.010
	4	0.562 $\pm$ 0.015	0.889 $\pm$ 0.005	0.265 $\pm$ 0.009	0.842 $\pm$ 0.008	0.945 $\pm$ 0.016	0.949 $\pm$ 0.008
	5	0.550 $\pm$ 0.013	0.890 $\pm$ 0.004	0.270 $\pm$ 0.005	0.841 $\pm$ 0.004	0.910 $\pm$ 0.019	0.952 $\pm$ 0.009
	6	0.522 $\pm$ 0.016	0.893 $\pm$ 0.004	0.203 $\pm$ 0.012	0.850 $\pm$ 0.005	0.934 $\pm$ 0.014	0.949 $\pm$ 0.014
	7	0.604 $\pm$ 0.018	0.893 $\pm$ 0.003	0.393 $\pm$ 0.018	0.840 $\pm$ 0.008	0.876 $\pm$ 0.023	0.962 $\pm$ 0.008
	8	0.607 $\pm$ 0.014	0.866 $\pm$ 0.003	0.388 $\pm$ 0.011	0.798 $\pm$ 0.008	0.891 $\pm$ 0.017	0.954 $\pm$ 0.007
	9	0.614 $\pm$ 0.011	0.874 $\pm$ 0.005	0.387 $\pm$ 0.008	0.821 $\pm$ 0.008	0.908 $\pm$ 0.013	0.943 $\pm$ 0.008
	10	0.602 $\pm$ 0.013	0.859 $\pm$ 0.008	0.360 $\pm$ 0.005	0.774 $\pm$ 0.010	0.916 $\pm$ 0.017	0.969 $\pm$ 0.010
	11	0.601 $\pm$ 0.012	0.864 $\pm$ 0.008	0.361 $\pm$ 0.007	0.792 $\pm$ 0.010	0.911 $\pm$ 0.020	0.956 $\pm$ 0.008
	12	0.640 $\pm$ 0.010	0.868 $\pm$ 0.006	0.416 $\pm$ 0.006	0.801 $\pm$ 0.007	0.929 $\pm$ 0.019	0.954 $\pm$ 0.006
	13	0.645 $\pm$ 0.011	0.862 $\pm$ 0.005	0.436 $\pm$ 0.005	0.793 $\pm$ 0.004	0.915 $\pm$ 0.018	0.951 $\pm$ 0.012
	14	0.654 $\pm$ 0.010	0.859 $\pm$ 0.006	0.479 $\pm$ 0.004	0.795 $\pm$ 0.008	0.880 $\pm$ 0.017	0.941 $\pm$ 0.011
	15	0.661 $\pm$ 0.009	0.870 $\pm$ 0.006	0.460 $\pm$ 0.009	0.814 $\pm$ 0.009	0.921 $\pm$ 0.014	0.941 $\pm$ 0.006
	16	0.685 $\pm$ 0.008	0.870 $\pm$ 0.007	0.477 $\pm$ 0.009	0.824 $\pm$ 0.008	0.953 $\pm$ 0.015	0.930 $\pm$ 0.008
	17	0.757 $\pm$ 0.006	0.886 $\pm$ 0.010	0.612 $\pm$ 0.016	0.841 $\pm$ 0.005	0.944 $\pm$ 0.010	0.943 $\pm$ 0.016
	18	0.781 $\pm$ 0.008	0.880 $\pm$ 0.008	0.636 $\pm$ 0.013	0.840 $\pm$ 0.002	0.967 $\pm$ 0.007	0.931 $\pm$ 0.016
	19	0.805 $\pm$ 0.012	0.877 $\pm$ 0.008	0.673 $\pm$ 0.020	0.831 $\pm$ 0.007	0.976 $\pm$ 0.003	0.935 $\pm$ 0.011
	20	0.794 $\pm$ 0.010	0.878 $\pm$ 0.006	0.650 $\pm$ 0.018	0.831 $\pm$ 0.003	0.979 $\pm$ 0.004	0.937 $\pm$ 0.011
	21	0.805 $\pm$ 0.010	0.871 $\pm$ 0.007	0.673 $\pm$ 0.016	0.831 $\pm$ 0.004	0.975 $\pm$ 0.006	0.924 $\pm$ 0.012
	22	0.871 $\pm$ 0.006	0.873 $\pm$ 0.006	0.811 $\pm$ 0.005	0.813 $\pm$ 0.007	0.949 $\pm$ 0.006	0.951 $\pm$ 0.007
	23	0.873 $\pm$ 0.007	0.885 $\pm$ 0.005	0.811 $\pm$ 0.006	0.835 $\pm$ 0.007	0.953 $\pm$ 0.007	0.950 $\pm$ 0.006
	24	0.848 $\pm$ 0.009	0.856 $\pm$ 0.004	0.749 $\pm$ 0.017	0.807 $\pm$ 0.006	0.975 $\pm$ 0.007	0.918 $\pm$ 0.009

H Reference Systems: the Fine-Tuning Ceilings

Tables 6 and 7 carry three reference rows below the rule, one supervised fine-tuning ceiling per fine-tuned model. Table 21 gives all three with the pre-trained rows they are compared against. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third answers the remaining question, whether supervision can clear the best cell the correction produces anywhere in the paper: KaLM-mini holds it, at 91.7 assignment accuracy and at 89.4 DirAcc@1 tied with Qwen3-0.6B. All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate 2×10^{-5} with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged. The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint is chosen by dev DirAcc@1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool’s whole resolution; without that, a model whose pre-trained encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of 6×10^{-4} , which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects $D = 10$, the top

of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and $D = 2$ at the layer its Task B row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose pre-trained encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its Task B row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches 4×10^{-5} on Qwen3-0.6B by epoch 4 and 3×10^{-5} on KaLM-mini, so the 499 pairs are memorized and more epochs buy nothing.

Lexical reference systems. Table 22 gives the three lexical reference systems (BM25, character 3–5-gram TF-IDF, and Levenshtein similarity) on both tasks; the character n-gram row is the reference quoted in Section 7.

The parameter-free repair matches contrastive fine-tuning on LaTa routing, but supervision clears it on Qwen3-0.6B and KaLM-mini. Fine-tuning LaTa on the positive train pairs (499 after a directory-level dev carve; setup above) gives an uncorrected Task A AUROC of 0.984, above every zero-shot baseline row (0.838 to 0.972), and 83.4 assignment accuracy; with ABTT on top it reaches 87.8 assignment accuracy and 85.2 DirAcc@1, below every zero-shot ABTT cell (88.5 to 91.7 and 86.1 to 89.4). ABTT also lowers that row’s AUROC from 0.984 to 0.970 while widening its cosine gap from 0.387 to 0.548, consistent with fine-tuning already removing part of the common component: the repair is not free on every row, and it matches what the available supervision buys on routing at this budget. The same recipe lands elsewhere on the other two encoders we fine-tuned: Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 DirAcc@1 with ABTT, and KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell. Over five Task B query and reference re-seedings the fine-tuned encoder with ABTT beats its zero-shot ABTT row by 1.8 points on Qwen3-0.6B (0.923 ± 0.002 against 0.905 ± 0.004) and by 2.1 points on KaLM-mini (0.930 ± 0.005 against 0.909 ± 0.006), while on LaTa the two are 0.877 and 0.877. So the match is a statement about LaTa, not about post-processing in general; it does not survive on the model with the best zero-shot cell in Table 7. All three ceilings are also flattered: 206 of the 535 test queries whose directory already ex-

System	Task A		Task B	
	AUROC	Cosine gap	Assign. acc.	Dir. acc.@1
LaTa (pre-trained)	0.938	0.237	73.8	72.1
LaTa (pre-trained) + ABTT	0.971	0.525	88.5	86.1
LaTa (fine-tuned)	0.984	0.387	83.4	81.6
LaTa (fine-tuned) + ABTT	0.970	0.548	87.8	85.2
Qwen3-0.6B (pre-trained)	0.966	0.024	82.3	80.3
Qwen3-0.6B (pre-trained) + ABTT	0.973	0.553	91.5	89.4
Qwen3-0.6B (fine-tuned)	0.996	0.684	91.4	90.8
Qwen3-0.6B (fine-tuned) + ABTT	0.994	0.716	92.1	91.4
KaLM-mini (pre-trained)	0.972	0.056	87.5	85.9
KaLM-mini (pre-trained) + ABTT	0.981	0.568	91.7	89.4
KaLM-mini (fine-tuned)	0.997	0.636	92.4	91.7
KaLM-mini (fine-tuned) + ABTT	0.994	0.651	93.4	92.5

Table 21: Supervised fine-tuning reference ceiling on LaTa, Qwen3-0.6B and KaLM-mini. The fine-tuned encoders are trained contrastively on 499 of the 565 positive train pairs (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of train by directory for model selection; the test split is untouched until this evaluation. Task A columns are test AUROC and cosine gap at the layer chosen by train AUROC; Task B columns are test assignment accuracy and directory accuracy at rank 1, in percent, at the layer chosen by train directory accuracy, with τ learned on train. The selected layers are listed in Table 9. ABTT rows for LaTa sweep D per layer on train and select $D = 10$ everywhere, which is the top of the grid $\{1, 2, 3, 5, 7, 10\}$, so the tuned and fixed variants coincide. The selected checkpoint for LaTa is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for Qwen3-0.6B sweep D per layer on train over the grid $\{1, 2, 3, 5, 7, 10\}$; 12 of 28 layer rows select the top of the grid. The selected checkpoint for Qwen3-0.6B is epoch 3 of 6 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for KaLM-mini sweep D per layer on train over the grid $\{1, 2, 3, 5, 7, 10\}$; 19 of 24 layer rows select the top of the grid. The selected checkpoint for KaLM-mini is epoch 1 of 4 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

ists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so read each margin as an upper bound.

I Token Attribution at Retrieval-Selected Layers

This appendix compares local explanations of the retrieval score before and after ABTT: methods, metrics, setup, and results. Appendix J gives the secondary metrics and sweeps, and Appendix K gives one qualitative example.

I.1 Local Attribution Methods

We evaluate local explanations for a fixed query-candidate retrieval score. The design goal is a review aid: attributions are shown as highlights connecting matching tokens across the two witnesses, so a scholar judging a proposed match can see which words the score rests on. Our metrics test faithfulness, whether a highlight tracks what the score depends on, not plausibility, whether it points at the words a philologist would compare (Jacovi and Goldberg, 2020). Integrated gradients

(Sundararajan et al., 2017) attributes the pairwise cosine to the input tokens of each side of the pair (pair-matrix view: Figure 8).

Our learned-mask attribution is adapted from MaRC (Brinner and Zarrieß, 2023). MaRC was proposed for classification, where a soft input mask preserves a target class score while sparsity and smoothness regularizers encourage concise rationales. We keep the per-instance mask optimization and replace the classification target with pairwise retrieval cosine: the mask is optimized on one side of a query-candidate pair while the partner representation stays fixed, and swapping the sides gives candidate-side attribution.

Let $\mathbf{E}(\mathbf{x}_q) \in \mathbb{R}^{T_q \times d_{\text{emb}}}$ denote the input-embedding lookup, $E_{>0}$ the encoder with the lookup removed, and $\mathbf{z}_c^*$ the candidate representation held fixed during optimization. We parameterize $\mathbf{m} = \sigma(\boldsymbol{\zeta})$ with free logits $\boldsymbol{\zeta} \in \mathbb{R}^{T_q}$ and write $\mathcal{I}$ for content-token positions. For the ABTT variant, define $s_{\text{ABTT}}(\mathbf{a}, \mathbf{b}) = \cos(\text{ABTT}(\mathbf{a}), \mathbf{b})$.

Baseline	Task A		Task B	
	AUROC	Gap	Assign.	Dir@1
BM25 (word)	0.974	0.243	62.5	56.3
TF-IDF char 3–5	0.987	0.555	91.8	89.9
Levenshtein	0.952	0.362	87.8	86.2

Table 22: Lexical baselines on the same 50/50 split, the same held-out test set and the same evaluation code as the embedding models, so the numbers are directly comparable to the headline Task A and Task B tables. Task A is measured over the test $n \times n$ score matrix: AUROC treats same-directory test pairs as positives, and Gap is the mean same-directory minus the mean different-directory score. Task B is reported in percent: Assign. is the existing-versus-new decision against a threshold τ fitted on train, and Dir@1 additionally requires the correct directory. TF-IDF char 3–5 is the strongest of the three at 89.9 percent Dir@1, which puts plain surface-form overlap within reach of the best embedding configurations on this corpus of hand-copied witnesses. BM25 separates pairs well but routes poorly: a raw BM25 score grows with query length, so each file’s maximum score tracks document length as much as relatedness, and the train-fitted τ transfers badly. BM25 uses $k_1 = 1.5$, $b = 0.75$ with document frequencies, vocabulary and average document length taken from train files only; the character TF-IDF vectoriser is likewise fitted on train only, and normalised Levenshtein has nothing to fit. Every score matrix is min-max rescaled using the train block; the map is monotone, so AUROC is unchanged, while the threshold τ is fitted on a fixed grid afterwards, which matters for BM25 (unrescaled, τ pins at the grid ceiling) and is the identity for char TF-IDF, whose train scores already span $[0, 1]$. Under the 5-seed query-versus-reference Task B protocol, in which only the query/reference partition of the test half is redrawn, directory accuracy at rank 1 is BM25 (word) 68.3 ± 7.1 ; TF-IDF char 3–5 91.1 ± 0.5 ; Levenshtein 87.2 ± 0.3 .

The retrieval-adapted MaRC loss is

$$\mathcal{L}(\mathbf{m}) = (s_{\text{ABTT}}(\tilde{\mathbf{z}}_q(\mathbf{m}), \mathbf{z}_c^*) - s_{\text{ABTT}}(\mathbf{z}_q, \mathbf{z}_c^*))^2 + \lambda \sum_{i \in \mathcal{I}} \frac{m_i}{|\mathcal{I}|} + \gamma \sum_{i, i+1 \in \mathcal{I}} \frac{|m_{i+1} - m_i|}{|\mathcal{I}| - 1}, \quad (1)$$

where $\tilde{\mathbf{z}}_q(\mathbf{m}) = \text{pool}(E_{>0}(\text{diag}(\mathbf{m}) \mathbf{E}(\mathbf{x}_q)), \mathbf{a}_q)$. The baseline variant uses raw cosine in place of the ABTT-corrected cosine in Equation 1. Logits outside $\mathcal{I}$ are frozen at $-\infty$.

I.2 Attribution Metrics

We report two threshold-free metrics: ρ_{LOO} , the Spearman correlation between attribution magnitude and the leave-one-out drop in the same retrieval score, and the DelAUC gap, the random-

order minus attribution-order deletion-curve area. A shuffled-attribution control recomputes each of six candidate metrics (ρ_{LOO} , Kendall τ_b , the DelAUC and InsAUC gaps, AOPC comprehensiveness and sufficiency) on five permutations of the attribution’s own scores in all twelve model-method-variant cells; a metric qualifies for the headline table (Table 23) only if the real attribution beats the permutation mean in every cell. ERASER-style sufficiency and comprehensiveness (DeYoung et al., 2020) and MinFrac, defined for retrieval cosine in Appendix J, stay out of the headline table because their verdicts depend on the threshold and on the erasure operator. Every curve metric ranks tokens by $|a|$, ties broken by position; Appendix J reports them and states both conventions and their consequences.

I.3 Setup and Layer Selection

MaRC runs up to 200 Adam steps per pair (learning rate 0.1, stopping after step 50 once the masked cosine is within 0.01 of the full one) from $\zeta_0 = \log 9$ with $\lambda = 0.01$ and $\gamma = 0.001$; integrated gradients uses 40 integration steps. Attribution layers follow a predeclared train-only rule, a variant of the Task B rule of Section 3: the earliest layer within 0.5 points of the best training-set DirAcc@1 under ABTT. It selects LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 for attribution, with $D = 10$ ABTT components; the most anisotropic layers on the test split, LaTa 8, PhilTa 6, and mT5-base 5, are kept as mechanism checks (Appendix B). The rule locks these layers before any attribution metric is computed.

I.4 Results

Table 23 compares baseline and ABTT attributions on 200 positive query-candidate pairs per model for LaTa, PhilTa, and mT5-base. ABTT repairs the scores being explained at these layers: LaTa layer 7 moves from AUROC 0.498 to 0.962 and DirAcc@1 30.2 to 86.8, PhilTa layer 1 from 0.939 to 0.977 and 69.3 to 88.3, and mT5-base layer 1 from 0.822 to 0.979 and 45.1 to 88.5.

ABTT improves rank faithfulness in five of six model-view cells (ρ_{LOO} 5/6, Kendall τ_b agreeing in the same five) by 10.3 to 22.5 standard errors, the sixth a tie at 1.5; chance-corrected deletion faithfulness improves in four of six, and both LaTa cells favour the baseline, at 2.1 and 8.5 standard errors. Comp@25% and the InsAUC gap point the same way (Table 24): Comp@25% favours ABTT

Model	Method	ρ_{LOO}		DeLAUC gap	
		base	ABTT	base	ABTT
LaTa	IG	0.003	0.370	0.859	0.181
	MaRC	0.083	0.538	0.524	0.327
PhilTa	IG	0.329	0.614	0.113	0.403
	MaRC	0.469	0.445	0.201	0.313
mT5-base	IG	0.143	0.686	0.061	0.561
	MaRC	0.257	0.504	0.042	0.463

Table 23: Attribution faithfulness at the predeclared operational layers, 200 positive pairs per model, for integrated gradients (IG) and retrieval-adapted MaRC. ρ_{LOO} correlates attribution magnitude with the leave-one-out change in the cosine. DeLAUC gap is the random-order minus attribution-order deletion-curve area, positive when attribution beats chance; the reference runs from 0.692 to 0.961 here, so zero is chance. Higher is better in both; boldface marks the better variant. ABTT wins 5/6 and 4/6. Ties, within 2 standard errors of zero: ρ for PhilTa MaRC. Ratio metrics are undefined below a full-query cosine of 0.05, so the DeLAUC columns average 195 to 199 ABTT pairs against 199 to 200 baseline pairs. Cross-variant comparisons are descriptive. Secondary metrics: Table 24.

in four of six cells, both LaTa cells to the baseline as for DeLAUC gap, and the InsaAUC gap in five of six, LaTa/IG to the baseline; MinFrac@0.80 does not, favouring ABTT in two. Neither is headline evidence, and InsaAUC gap and AOPC-Suff (identical gaps by construction) fall to -0.024 on mT5-base/IG baseline, the one non-positive cell of the control (Table 25). So we claim better-ranked rationales, not a gain on every axis.

Masking acts on cached layer- L hidden states, so every faithfulness statement is about the pooled read-out at that layer, not the encoder’s input-to-output map. Appendix J carries the threshold sweeps and Figure 7; Appendix K shows one PhilTa retrieval case, with its integrated-gradient pair matrix before and after ABTT.

J Secondary Attribution Metrics and Sweeps

Ranking conventions. Two ranking conventions hold for every curve metric and every method, variant and control alike. Tokens are ranked by attribution magnitude $|a|$ rather than by signed importance, because the question is which tokens carry the pair cosine and both methods produce two-signed scores. Ties are broken by token position, which matters most for MaRC, whose many exactly-zero scores are then ordered by reading order. Absolute values are therefore not comparable

with a signed-ranking implementation, and Kendall τ_b in Table 24 is the tie-corrected companion to ρ_{LOO} under the second convention.

Secondary metrics. Figure 7 plots the rank-faithfulness column of Table 23. Table 24 carries the metrics Table 23 does not: tie-corrected Kendall τ_b , chance-corrected insertion faithfulness, and the three ERASER-style rationale metrics at the global threshold settings. Sufficiency keeps the top-ranked query tokens and measures how much of the full pair cosine remains, comprehensiveness removes those tokens and measures the score drop, and MinFrac reports the smallest retained query-token fraction needed to recover a fixed share of the full cosine. Four computed quantities are printed in neither table: the raw deletion and insertion areas, which appear here only as the chance-corrected gaps, and AOPC-Sufficiency and AOPC-Comprehensiveness, whose per-threshold components are in the sweeps below. Table 25 gives the per-cell shuffled-attribution gaps for the six candidate metrics, the control Appendix I quotes as a validity statement. The rest of this appendix expands those rationale metrics across Sufficiency and Comprehensiveness thresholds of 10%, 25%, and 50%, and MinFrac recovery thresholds of 0.70, 0.80, 0.90, and 0.95. We use one global midpoint setting rather than selecting thresholds per model or per method. Table 26 gives the sweep for integrated gradients and retrieval-adapted MaRC, and Table 27 gives it for the supplemental attribution methods. Every table in this appendix uses the same representation-level erasure operator as Table 23, so the cells are comparable across them.

K Qualitative Attribution Examples

This appendix keeps the example-level views for one PhilTa retrieval pair. Figure 8 gives the integrated-gradient pair matrix before and after ABTT. The attention map in Figure 9 is context only, since ABTT is applied after encoding. The retrieval-adapted MaRC masks in Figure 10 show the learned-mask view for the same pair.

L Cluster Geometry

Figures 11 and 12 give the t-SNE and UMAP panels for LaTa, PhilTa, and mT5-base, and Figures 13 and 14 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at $D = 10$;

Model	Method	τ_{LOO}		InsAUC gap		Suff@25%		Comp@25%		MinFrac@0.80	
		base	ABTT	base	ABTT	base	ABTT	base	ABTT	base	ABTT
LaTa	IG	0.007	0.262	0.284	0.061	0.990	0.804	1.242	0.177	0.111	0.272
	MaRC	0.068	0.400	0.088	0.138	0.731	0.917	0.781	0.280	0.184	0.177
PhilTa	IG	0.234	0.462	0.004	0.245	0.814	0.971	0.113	0.522	0.240	0.136
	MaRC	0.334	0.318	0.063	0.193	0.888	0.878	0.093	0.363	0.149	0.259
mT5-base	IG	0.101	0.515	-0.024	0.399	0.933	1.126	0.012	0.642	0.086	0.178
	MaRC	0.181	0.359	0.003	0.327	0.960	1.013	0.014	0.582	0.031	0.246

Table 24: Secondary attribution metrics, on the same pairs, the same layers and the same erasure operator as Table 23. Boldface marks the better variant within a pair; higher is better everywhere except MinFrac. None of these columns is in the main table, and each is out for its own reason. Kendall τ_b agrees with ρ_{LOO} in 5/6 cells, but it is the tie-corrected twin of the same statistic rather than a second witness. Chance-corrected insertion faithfulness favours ABTT in 5/6 cells, and we do not report it in the main table because in one of the twelve cells, a baseline cell, the real attribution does not beat a permutation of its own scores (Table 25), so the measurement does not meet the validity bar we set for a headline column. The threshold-based ERASER metrics are reported for completeness: their baseline-versus-ABTT verdict depends on the threshold and on the erasure operator, which is why the main table uses threshold-free, chance-corrected metrics instead. The InsAUC columns average 195 to 199 ABTT pairs against the baseline’s 199 to 200, for the same small-denominator reason. Full threshold sweeps are in Tables 26 and 27.

Model	Method	Variant	ρ_{LOO}	τ_{LOO}	DelAUC gap, AOPC-Comp	InsAUC gap, AOPC-Suff
LaTa	IG	base	+0.007 (0.018)	+0.010 (0.013)	+0.887 (0.071)	+0.300 (0.044)
		ABTT	+0.379 (0.014)	+0.268 (0.010)	+0.183 (0.009)	+0.064 (0.008)
	MaRC	base	+0.087 (0.021)	+0.070 (0.016)	+0.509 (0.082)	+0.073 (0.072)
		ABTT	+0.535 (0.018)	+0.398 (0.014)	+0.327 (0.010)	+0.140 (0.011)
PhilTa	IG	base	+0.327 (0.017)	+0.233 (0.012)	+0.113 (0.006)	+0.006 (0.003)
		ABTT	+0.614 (0.013)	+0.462 (0.010)	+0.406 (0.011)	+0.246 (0.010)
	MaRC	base	+0.466 (0.016)	+0.332 (0.012)	+0.202 (0.005)	+0.065 (0.004)
		ABTT	+0.446 (0.019)	+0.319 (0.014)	+0.316 (0.015)	+0.195 (0.016)
mT5-base	IG	base	+0.143 (0.025)	+0.101 (0.018)	+0.061 (0.003)	-0.024 (0.001)
		ABTT	+0.688 (0.013)	+0.517 (0.011)	+0.556 (0.020)	+0.397 (0.026)
	MaRC	base	+0.256 (0.021)	+0.180 (0.015)	+0.042 (0.001)	+0.003 (0.001)
		ABTT	+0.502 (0.017)	+0.357 (0.013)	+0.460 (0.022)	+0.323 (0.028)

Table 25: Shuffled-attribution control for the six candidate metrics, on the same pairs, layers and erasure operator as Table 23. Each entry is the real metric minus its mean over five permutations of the same attribution’s scores across the query tokens, with the standard error over pairs in parentheses: a positive gap means the real attribution beats a fake one drawn from its own score distribution, and a metric passes the control only when every cell is positive. DelAUC gap and AOPC-Comprehensiveness share one column, as do InsAUC gap and AOPC-Sufficiency, because the random-order reference and the constant trapezoid offset cancel in the difference, so each pair has the same gap to machine precision and one verdict. Boldface marks the one cell at or below zero: mT5-base IG baseline on InsAUC gap, at -0.024. Four further cells are positive but within two standard errors of zero: LaTa IG baseline on ρ_{LOO} , LaTa IG baseline on τ_{LOO} , LaTa MaRC baseline on InsAUC gap and PhilTa IG baseline on InsAUC gap. The rank columns use 200 pairs per cell; the AUC columns, undefined below a full-query cosine of 0.05, average 195 to 199 ABTT pairs against 199 to 200 baseline pairs.

each model sits at the layer chosen on the train split by DirAcc@1 under ABTT, the Task B ABTT layer of Table 9.

In t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) projections at the train-selected layers, ABTT-only with $D = 10$ forms tighter same-directory groups than baseline mean pooling: the t-SNE silhouette of the 1,160 winnable witnesses rises by 0.07 to 1.16 on the three T5 encoders.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	IG	0.003	0.370	0.949	0.556	0.990	0.804	1.014	0.969	1.144	0.069	1.242	0.177	1.203	0.354	0.082	0.196	0.111	0.272	0.159	0.388	0.197	0.475
	MaRC	0.083	0.538	0.696	0.797	0.731	0.917	0.807	1.004	0.487	0.112	0.781	0.280	0.889	0.556	0.139	0.101	0.184	0.177	0.270	0.328	0.317	0.430
PhiTa	IG	0.329	0.614	0.637	0.844	0.814	0.971	0.937	1.038	0.042	0.288	0.113	0.522	0.231	0.746	0.156	0.084	0.240	0.136	0.411	0.221	0.548	0.290
	MaRC	0.469	0.445	0.786	0.691	0.888	0.878	0.971	1.001	0.032	0.169	0.093	0.363	0.263	0.625	0.076	0.199	0.149	0.259	0.299	0.350	0.435	0.406
mT5-base	IG	0.143	0.686	0.839	0.903	0.933	1.126	0.976	1.210	0.004	0.312	0.012	0.642	0.038	1.005	0.045	0.126	0.086	0.178	0.184	0.252	0.323	0.304
	MaRC	0.257	0.504	0.918	0.791	0.960	1.013	0.989	1.121	0.004	0.274	0.014	0.582	0.042	0.902	0.016	0.175	0.031	0.246	0.085	0.347	0.199	0.412

Table 26: Appendix attribution sweep for the two paper-facing attribution views, IG and retrieval-adapted MaRC, across the three main-paper models. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% $\uparrow$		Suff@25% $\uparrow$		Suff@50% $\uparrow$		Comp@10% $\uparrow$		Comp@25% $\uparrow$		Comp@50% $\uparrow$		MinFrac@0.70 $\downarrow$		MinFrac@0.80 $\downarrow$		MinFrac@0.90 $\downarrow$		MinFrac@0.95 $\downarrow$	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	BERTScore	0.282	0.496	-0.019	0.565	-0.032	0.788	0.016	0.960	0.292	0.074	0.298	0.192	0.282	0.426	0.377	0.210	0.403	0.297	0.436	0.433	0.467	0.517
	OT	0.031	0.382	0.929	0.581	0.961	0.820	0.979	0.974	0.872	0.071	0.882	0.179	0.827	0.361	0.087	0.184	0.115	0.258	0.160	0.376	0.197	0.464
	Att-weighted	-0.014	0.001	-0.185	0.265	-0.216	0.459	0.341	0.709	0.015	0.046	0.022	0.135	0.182	0.255	0.334	0.476	0.347	0.584	0.377	0.705	0.408	0.787
	Att-standalone	0.144	0.575	0.788	0.721	0.910	0.905	0.962	1.010	0.737	0.108	0.954	0.275	1.074	0.553	0.116	0.132	0.172	0.209	0.273	0.341	0.322	0.436
	DLA	0.075	0.601	0.973	0.704	1.009	0.893	1.035	0.989	1.200	0.123	1.250	0.292	1.251	0.555	0.086	0.128	0.118	0.208	0.154	0.353	0.192	0.452
	random	0.004	-0.002	0.265	0.547	0.519	0.736	0.819	0.871	0.026	0.021	0.030	0.053	0.177	0.136	0.183	0.220	0.219	0.330	0.288	0.514	0.338	0.659
	inverse	-0.003	-0.370	-0.202	0.276	-0.179	0.436	-0.205	0.653	-0.007	-0.017	-0.013	-0.022	-0.015	0.035	0.615	0.513	0.706	0.645	0.811	0.797	0.847	0.883
PhiTa	BERTScore	0.561	0.646	0.695	0.841	0.874	1.016	0.984	1.105	0.047	0.231	0.113	0.442	0.253	0.686	0.130	0.122	0.199	0.178	0.311	0.254	0.393	0.310
	OT	0.340	0.616	0.641	0.846	0.818	0.971	0.940	1.040	0.043	0.289	0.114	0.523	0.233	0.749	0.152	0.084	0.237	0.136	0.401	0.221	0.541	0.289
	Att-weighted	0.111	0.041	0.589	0.355	0.807	0.592	0.923	0.775	0.013	0.050	0.051	0.156	0.145	0.311	0.187	0.376	0.268	0.485	0.415	0.630	0.550	0.723
	Att-standalone	0.287	0.217	0.666	0.486	0.822	0.705	0.937	0.888	0.022	0.095	0.073	0.240	0.228	0.458	0.148	0.270	0.240	0.358	0.405	0.479	0.533	0.560
	DLA	0.531	0.599	0.729	0.872	0.862	0.970	0.959	1.006	0.068	0.311	0.159	0.567	0.337	0.776	0.102	0.069	0.182	0.115	0.345	0.205	0.486	0.285
	random	-0.004	-0.002	0.650	0.400	0.815	0.603	0.922	0.792	0.011	0.034	0.031	0.093	0.084	0.220	0.150	0.351	0.242	0.478	0.412	0.642	0.576	0.751
	inverse	-0.329	-0.614	0.514	0.046	0.636	0.109	0.774	0.255	0.002	-0.030	0.015	-0.049	0.065	-0.038	0.325	0.859	0.516	0.909	0.730	0.951	0.837	0.970
mT5-base	BERTScore	0.351	0.569	0.906	0.777	0.968	1.059	0.991	1.159	0.004	0.227	0.012	0.486	0.036	0.790	0.030	0.178	0.051	0.244	0.104	0.315	0.187	0.356
	OT	0.151	0.678	0.840	0.908	0.933	1.127	0.976	1.205	0.004	0.313	0.012	0.641	0.039	0.996	0.045	0.127	0.086	0.178	0.183	0.252	0.321	0.305
	Att-weighted	0.242	0.564	0.937	0.962	0.974	1.067	0.990	1.075	0.004	0.319	0.011	0.589	0.035	0.845	0.015	0.085	0.028	0.137	0.068	0.228	0.148	0.292
	Att-standalone	0.145	0.268	0.886	0.616	0.960	0.760	0.986	0.857	0.004	0.221	0.011	0.412	0.035	0.587	0.032	0.270	0.060	0.374	0.124	0.499	0.220	0.586
	DLA	0.237	0.552	0.901	0.934	0.950	1.051	0.979	1.032	0.004	0.341	0.014	0.622	0.046	0.825	0.024	0.083	0.045	0.133	0.110	0.231	0.245	0.312
	random	0.002	0.002	0.917	0.350	0.968	0.534	0.989	0.741	0.001	0.048	0.004	0.119	0.011	0.268	0.028	0.401	0.046	0.537	0.095	0.697	0.175	0.785
	inverse	-0.143	-0.686	0.687	-0.201	0.884	-0.186	0.962	0.002	0.002	-0.092	0.007	-0.168	0.025	-0.210	0.114	0.869	0.158	0.912	0.260	0.953	0.408	0.974

Table 27: Supplemental attribution sweep for additional methods and sanity baselines. *random* and *inverse* are diagnostic baselines, not paper-facing attribution methods. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

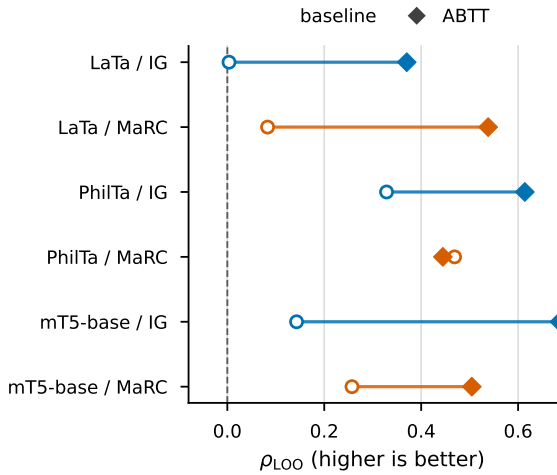


Figure 7: Leave-one-out rank correlation ρ_{LOO} at the predeclared operational attribution layers, for integrated gradients (IG) and retrieval-adapted MaRC. Each line connects the baseline and ABTT variants of one model-method cell. ABTT raises ρ_{LOO} in five of the six cells; the sixth, PhiTa MaRC, moves from 0.469 to 0.445, a tie within two standard errors (Table 23). Secondary metrics: Table 24.

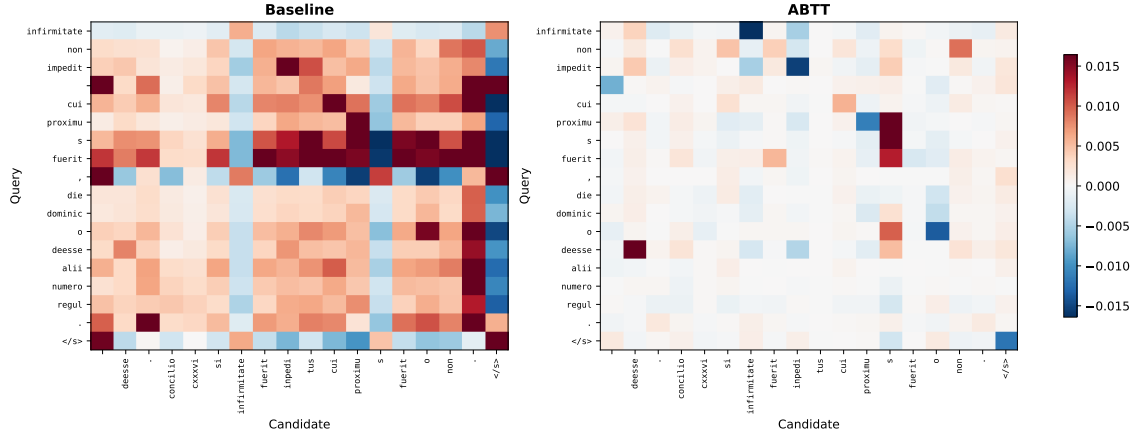


Figure 8: Derived pairwise heatmap for one PhilTa retrieval case. The left panel shows the baseline representation and the right panel shows the same example after ABTT. Both axes use original decoded tokens. Cell values combine token-token cosine similarity with single-sequence integrated-gradient scores.

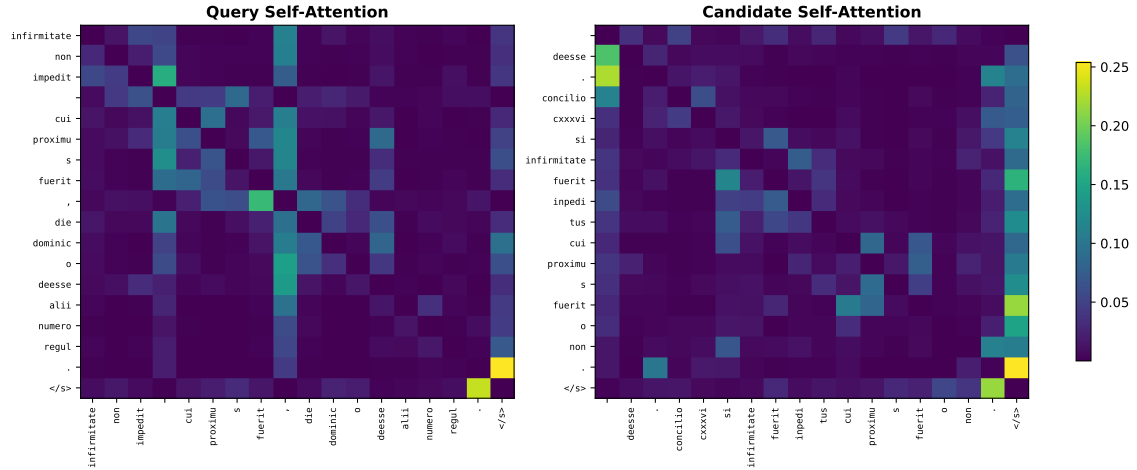


Figure 9: Companion self-attention maps for the same PhilTa example as Figure 8. ABTT is applied after encoding, so attention is included only as supporting context.

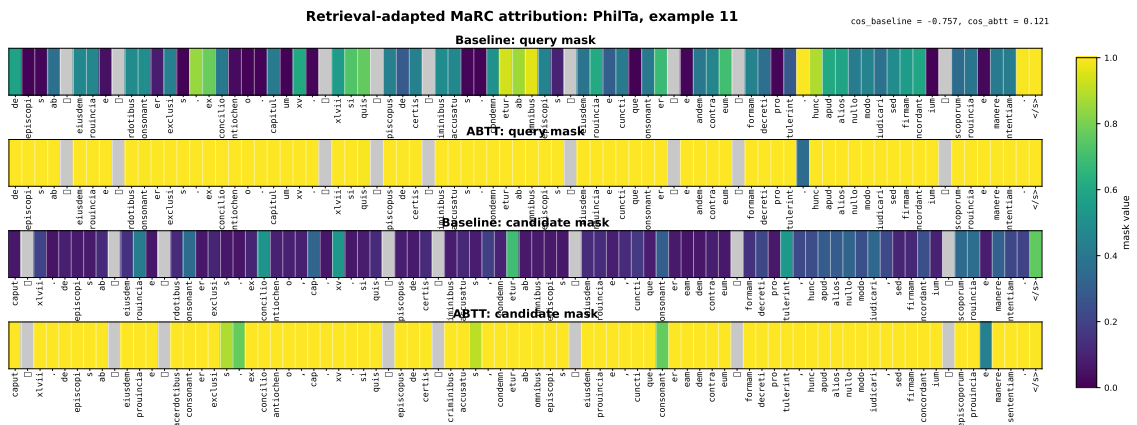


Figure 10: Retrieval-adapted MaRC masks for the same PhilTa pair as Figure 8. Top row: mask on the query with the candidate fixed. Bottom row: mask on the candidate with the query fixed. Left column: baseline cosine target. Right column: ABTT-corrected cosine target.

2D TSNE of labelled fragments at each model's train-selected layer

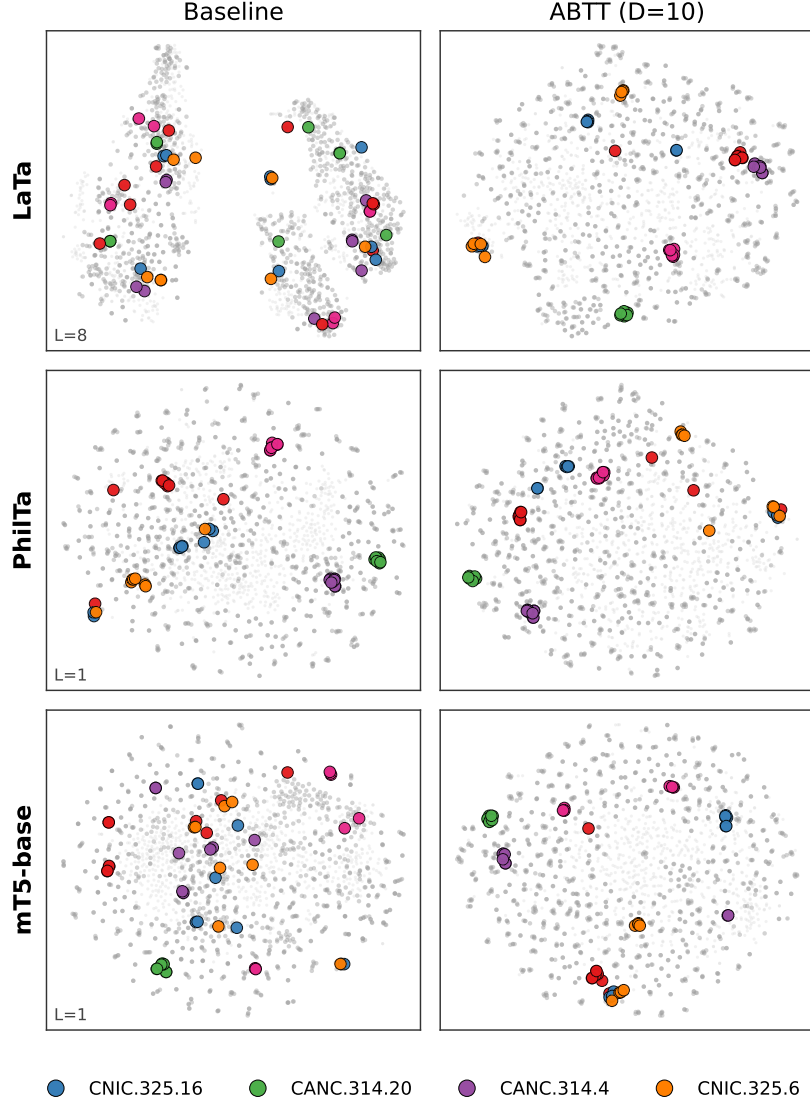


Figure 11: t-SNE projections of all 1,705 manuscript witnesses for LaTa, PhilTa, and mT5-base, each at the layer chosen on the train split: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel as L and equal to the Task B ABTT layer in Table 9. Columns compare baseline mean pooling with ABTT-only using $D = 10$, fit on the training embeddings. Colors mark the six largest directories; other witnesses of directories with at least two witnesses are mid gray, and singletons light gray. The silhouette scores in Appendix L are computed in these two-dimensional coordinates over the 1,160 winnable witnesses, those whose directory has at least two witnesses.

2D UMAP of labelled fragments at each model's train-selected layer

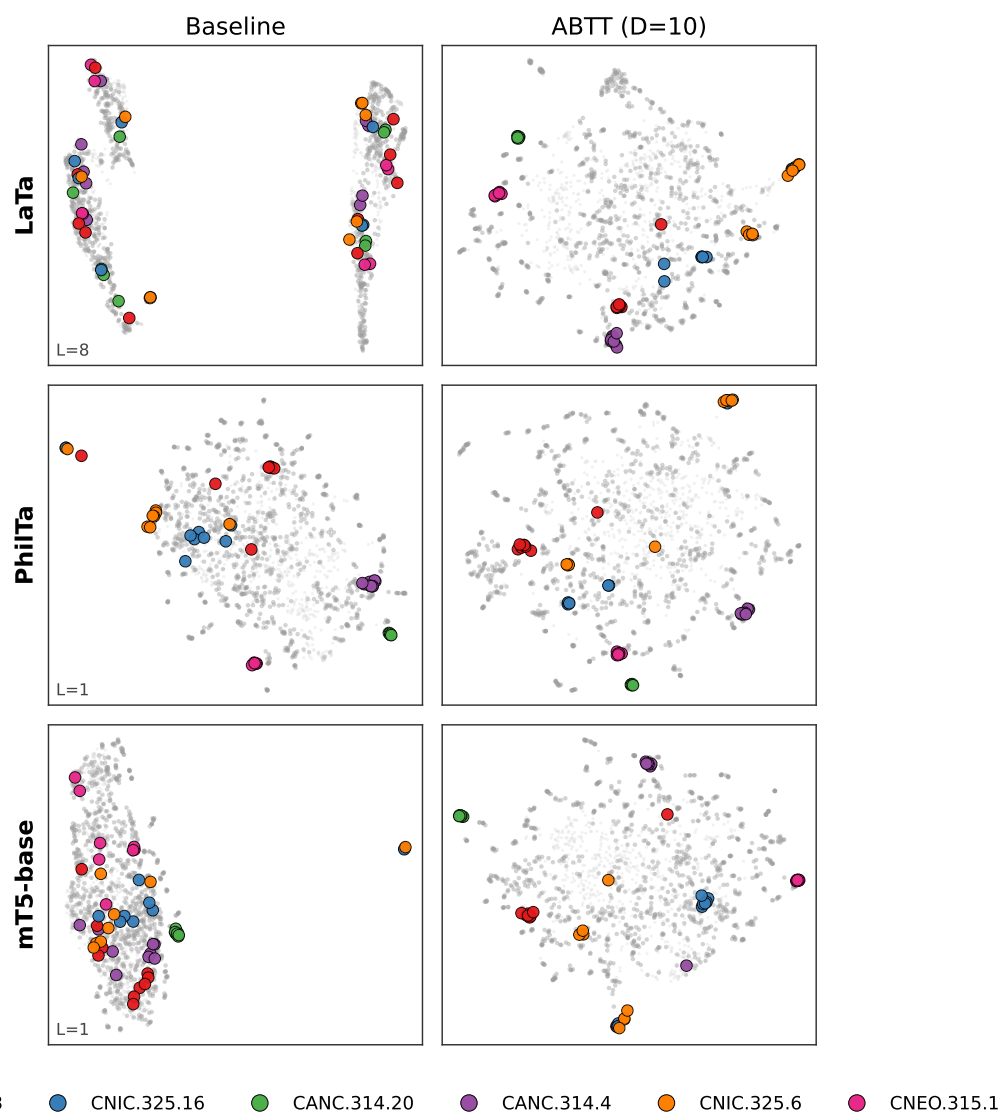


Figure 12: UMAP projections with the same configuration as Figure 11. The same cluster-tightening pattern appears under a different projection method.

2D TSNE of labelled fragments at each model's train-selected layer

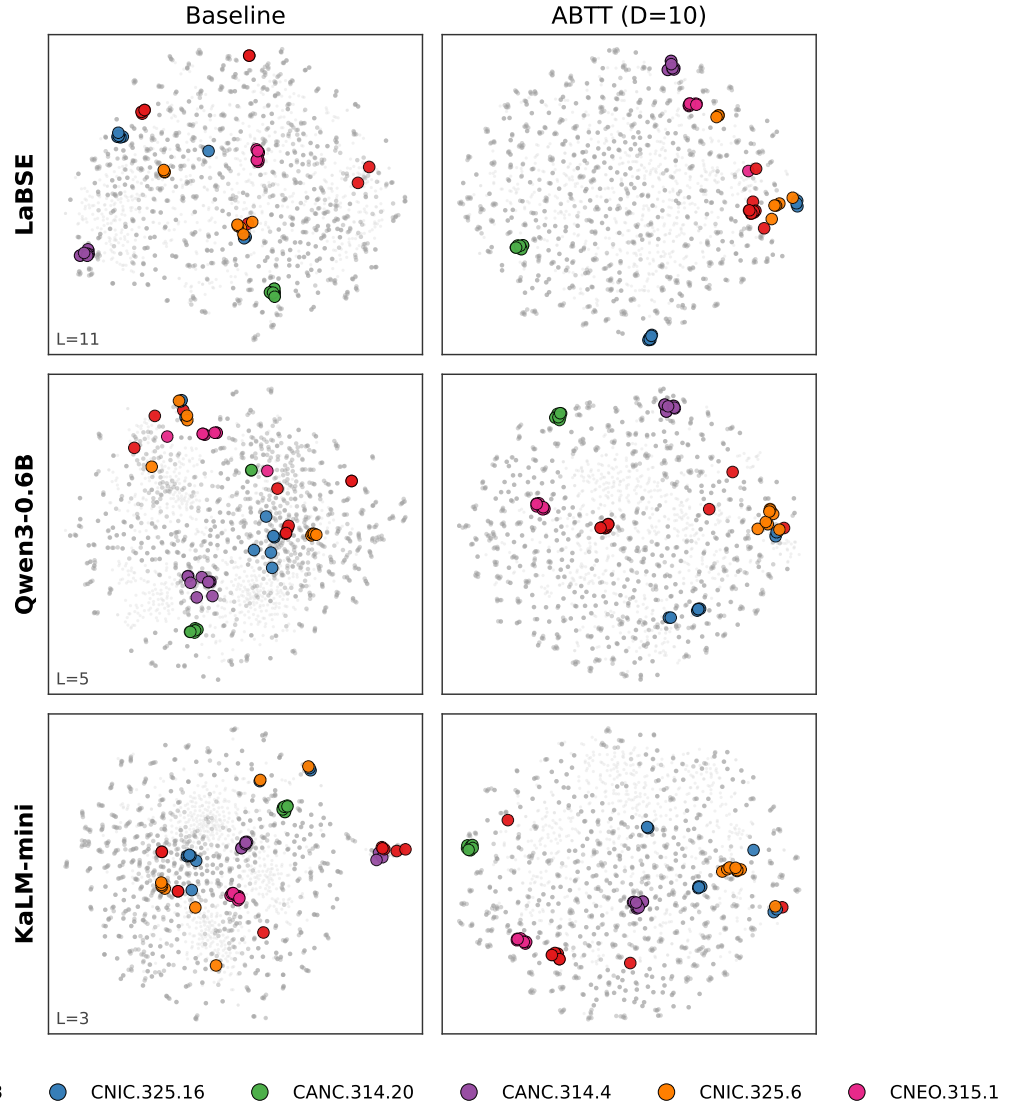


Figure 13: t-SNE projections of all 1,705 manuscript witnesses for LaBSE, Qwen3-0.6B, and KaLM-mini under the layer rule of Figure 11: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel and equal to the Task B ABTT layer in Table 9. Columns compare baseline mean pooling with ABTT-only using $D = 10$. Colors mark the same six largest directories as Figure 11.

2D UMAP of labelled fragments at each model's train-selected layer

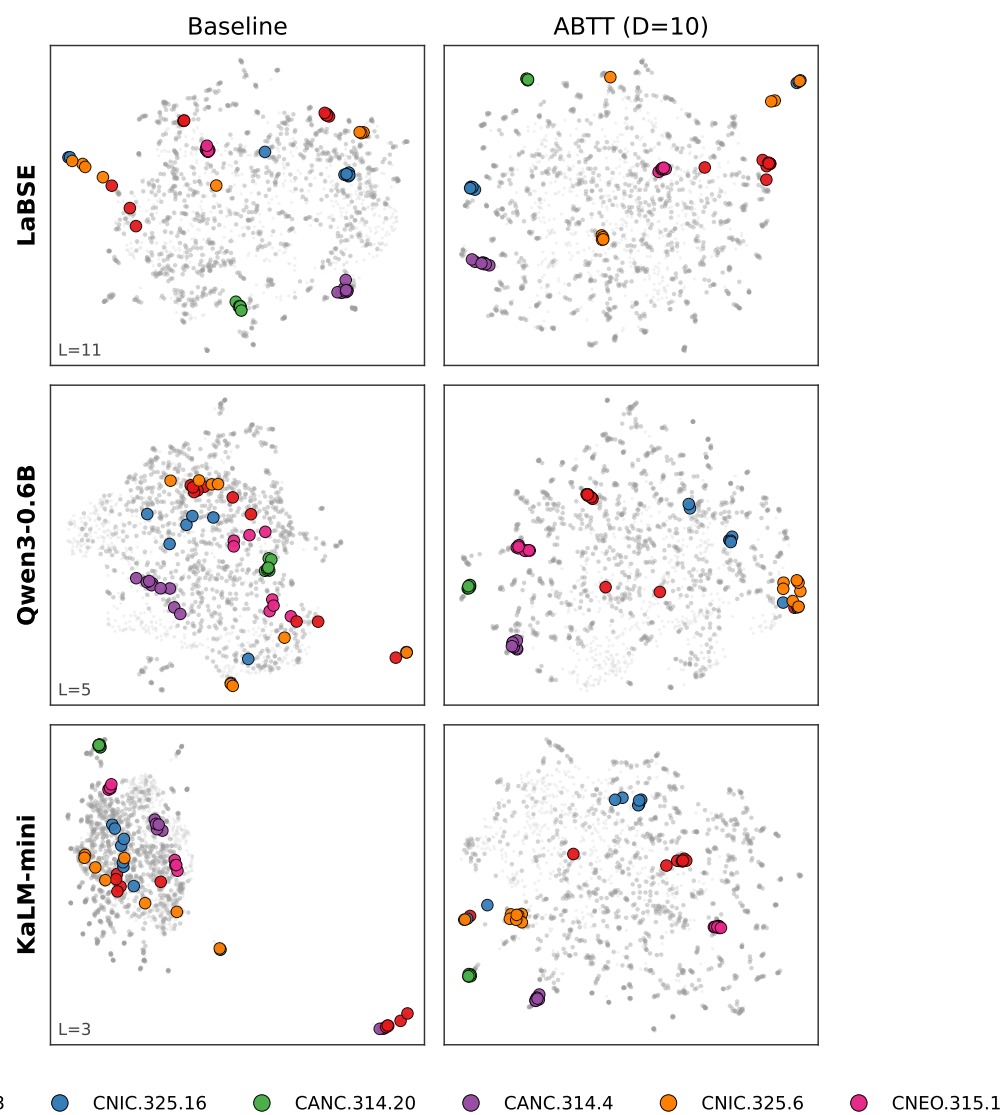


Figure 14: UMAP projections matching Figure 13. The cluster-tightening pattern appears under a second projection method.